



Numerical analysis of the prototype of the high-temperature thermal energy storage based on sand bed

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ARTICLE INFO

Handling editor: Henrik Lund

Keywords:

Thermal energy storage
TES
Sensible heat storage
High-temperature heat storage
Seasonal heat storage
Sand bed
CFD

ABSTRACT

Heating accounts for more than 60 % of the final energy consumed by buildings in EU countries. As new buildings are required to meet a zero-carbon standard, technical solutions are being sought to reduce heat demand and maximize the use of renewable energy. One of the possible solutions is the implementation of seasonal heat storage systems that can be charged using solar energy. The main goal of the study was to analyze the operation of a long-term, high-temperature heat storage using a sand bed and distinguish the parameters affecting its efficiency. In addition, the work aimed to show the feasibility of using such a storage in different charging schedules and the method of discharge through an air heat exchanger. As part of the research, a laboratory prototype consisting of an insulated tank filled with 300 kg of sand was developed. This prototype was used to perform the experiments required to validate the numerical model. The model, created in the ANSYS Workbench 2024 R2 environment, was then used to perform further analysis aimed at increasing the efficiency of the tested system. The results showed that the low thermal conductivity of sand prolonged high temperature retention and minimised heat losses, especially with thicker insulation. Analysis showed that positioning the heaters closer to the tank axis reduced heat loss, but required monitoring to avoid temperatures above ~1073 K. In addition, control logic was implemented to prevent the heaters from overheating the sand. Various charging schemes - continuous, off-peak G12 and photovoltaic based - demonstrated the flexibility of the system under changing energy price tariffs and renewable availability. Discharge tests used airflow in a pipe heat exchanger, where higher velocity increased power output but caused faster local cooling. The proposed solution could be cost-effective, environmentally friendly and suitable for both building heating and industrial processes, highlighting its potential for scalable, long-term energy storage.

1. Introduction

With the rapid growth in the utilization of renewable energy sources and the increasing challenges of integrating them stably into the electricity system, efficient energy storage methods are becoming increasingly crucial [1]. One way to solve this problem is to use different electricity storage systems, such as batteries [2], supercapacitors [3], compressed air energy storage (CAES) [4], or pumped hydro storage [5]. When there is a surplus of electricity production, the storage systems charge, and when demand arises, they discharge. Another alternative is storing energy in thermal energy storage (TES) devices. The most

common TESs uses an energy carrier whose temperature rises or whose state of aggregation changes, thereby storing heat. Examples include water reservoirs [6], seasonal underground heat storage [7], reversible chemical reactions [8], sensible heat storage in accumulative solid materials [9], phase change materials (PCMs) [10], molten salts [11], and sensible-latent heat thermal energy storage [12]. Each has different characteristics regarding the amount of heat stored, temperature, or charge/discharge rate.

The storage medium used for the TES systems significantly influences the performance of the storage systems. The most common components of sand are silica in the form of quartz, feldspar, and other minerals like carbonates, micas, amphiboles, and pyroxenes [13]. Sand is

This article is part of a special issue entitled: ICET 2024 published in Energy.

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<https://doi.org/10.1016/j.energy.2025.137472>

Received 28 February 2025; Received in revised form 8 May 2025; Accepted 7 July 2025

Available online 9 July 2025

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Symbols and abbreviations*Symbols*

ρ –	density $\left[\frac{\text{kg}}{\text{m}^3}\right]$
t –	time [s]
T –	temperature [K]
k_c –	thermal conductivity $\left[\frac{\text{W}}{\text{m}\cdot\text{K}}\right]$
C_p –	specific heat $\left[\frac{\text{J}}{\text{kg}\cdot\text{K}}\right]$
h_c –	heat transfer coefficient $\left[\frac{\text{W}}{\text{m}^2\cdot\text{K}}\right]$
Nu –	Nusselt number [–]
L –	characteristic length [m]
Ra –	Rayleigh number [–]
Pr –	Prandtl number [–]
θ –	inclination angle of the enclosure [°]
g –	gravitational acceleration $\left[\frac{\text{m}}{\text{s}^2}\right]$
β –	thermal expansion coefficient $\left[\frac{1}{\text{K}}\right]$
ν –	kinematic viscosity $\left[\frac{\text{m}^2}{\text{s}}\right]$
α –	thermal diffusivity $\left[\frac{\text{m}^2}{\text{s}}\right]$
μ –	dynamic viscosity [Pa·s]
u, v, w –	velocity for x, y, z coordinates $\left[\frac{\text{m}}{\text{s}}\right]$

p –	pressure [Pa]
x, y, z –	directional coordinates [m]
k –	turbulence kinematic energy $\left[\frac{\text{m}^2}{\text{s}^2}\right]$
ε –	turbulence dispersion rate $\left[\frac{\text{m}^2}{\text{s}^3}\right]$
ν_t –	turbulent viscosity $\left[\frac{\text{m}^2}{\text{s}}\right]$
$\sigma_\varepsilon, \sigma_k$ –	turbulent Prandtl numbers [–]
G_k –	generation of turbulence kinetic energy $\left[\frac{\text{kg}}{\text{m}\cdot\text{s}^3}\right]$
P –	power [W]
E –	energy [J]

Abbreviations

CAES	Compressed Air Energy Storage
CFD	Computational Fluid Dynamics
CPCM	Composite Phase Change Material
FEM	Finite Element Method
PCM	Phase Change Material
PLC	Programmable Logic Controller
SR	Storage Ratio
TES	Thermal Energy Storage
Wb	ANSYS Workbench
WFS-CPCM	Waste Foundry Sand Composite Phase Change Material
WHR	Waste Heat Recovery

characterized by.

- high thermal capacity – it can absorb and store substantial thermal energy, enhancing its effectiveness in energy storage applications,
- high thermal conductivity – it facilitates rapid heat transfer, enabling efficient charging and discharging of thermal energy,
- cost-effectiveness – it is an abundant, inexpensive material that requires no complex manufacturing processes, making it a readily available and economical option,
- long-term stability – it can withstand high temperatures without significant degradation, phase changes, or chemical reactions during thermal cycling,
- non-toxicity and environmental friendliness,
- compatibility with high-temperature applications [14].

Research into the use of solid heat storage has been ongoing for many years. One of the more widely described solutions using sand for energy storage and transport in high-temperature solar receivers was described in Ref. [15]. The discussed system avoids the use of gaseous and liquid heat transfer media. Studies have shown that sand can be heated to around 1000 °C, effectively absorbing solar radiation and allowing direct heat transfer to industrial systems. The disadvantage of such high-temperature heat storage is the sintering of the sand, which alters its properties and can impact the system's efficiency. The authors concluded that the focus should be on selecting thermal shock-resistant materials and developing methods to limit the degradation of the heat storage medium in long-term operation. Diago et al. [16] investigated the physical and chemical properties of seven sand samples from different locations in the United Arab Emirates. It was shown that the high calcium content of the sand and temperatures above 650–1000 °C can cause heat flow problems within the storage. It has also been demonstrated that the specific heat of the sand increases as its temperature rises. It should also be noted that compared to molten salt heat

storage, sand storage saves 90 % of the cost of the material itself, and the operating temperatures of such systems are much higher.

Different types of filling materials can be used for sand thermal storage. One is waste foundry sand, which cannot be 100 % recycled due to its poor post-process quality. This results in a significant increase in the metallurgical process cost but proves to be a relatively good TES fill material. Yousef et al. [17] presented the possibility of mixing waste foundry sand with sodium nitrate (NaNO_3) to produce composite phase change materials (CPCM). Such material has a temperature stability of up to 400 °C, with a heat capacity of 628 ± 27 kJ/kg and an average thermal conductivity of 1.38 W/(mK). This demonstrates that it is possible to reuse waste sand for medium-to-high-temperature thermal storage applications. Another option could be using sand or other bulk materials as energy storage in the form of potential energy [18]. In periods with an energy surplus from solar radiation, it is possible to utilize this energy to lift sand or other bulk materials, and then use it for energy production during periods of increased energy demand. In the case of sand, it was possible to obtain 247 kW of electricity with 56 % efficiency. Sand can be used to increase the efficiency of solar installations. Tei et al. [19] presented the possibility of increasing the energy yield of a flat plate solar collector by applying a layer of sand to the collector's absorber. This modified design allowed an increase in efficiency of about 32 % compared to an absorber without sand. Furthermore, to increase the thermal conductivity of the sand bed in TES, the sand could be doped with other materials to improve heat transfer. In this context, Tetteh et al. [20] used different metals to increase the heat flow. The best results were obtained with aluminium chips, increasing the thermal conductivity of the entire storage by 1.7 times that of pure sand. On the other hand, Ahmad et al. [21] demonstrated the potential of waste foundry sand composite phase change materials (WFS-CPCMs) for TES and waste heat recovery (WHR). The results demonstrate that a packed bed latent heat storage unit utilizing WFS-CPCMs can efficiently capture and reuse waste heat from industrial processes, particularly within the

Table 1

Selected studies carried out on the testing and development of high temperature thermal energy storage systems.

Objectives of the study	Materials used in the study	Findings	Reference
Evaluation of desert dune sand as thermal energy storage, heat transfer, and direct solar absorber material	Seven sand samples from the desert of the United Arab Emirates	Operation of sand-based thermal energy storage has been demonstrated up to 1000 °C, despite an initial mass loss during the first heating cycle	Diago et al. [16]
Assessment of the thermal performance of silica sand as a heat storage medium in a shell-and-tube sensible heat storage thermal energy system operating with water as the heat transfer fluid	Two types of silica sand, including fine sand and coarse sand	The fine sand, which had smaller particles compared to the coarse sand, was found to enhance the heat transfer in the system	Nikšar et al. [31]
Verification of a sand bed seasonal solar thermal storage concept requiring additional controls for residential heating applications	Silica sand	Seasonal sand-bed solar thermal storage systems could be a viable option for storing heat in regions with climates characterized by long periods of freezing temperatures	Hailu et al. [32]
Investigation of the influence of coil pitch, inlet fluid temperature, and hot fluid velocity on sand-based thermal energy storage unit	Two common sand types in the United Arab Emirates	The results showed that the three factors, including fluid inlet temperature, velocity, and number of coil turns, significantly affect the thermal energy stored	Bashria et al. [14]
Development of low-cost, simple, and scalable black coatings on solid particles with high solar absorptivity and demonstration of a simple and scalable method to convert low-cost quartz sand into a thermal energy storage material	Silica sand coated with silica shell containing black spinel nanoparticles	Coating low-cost silica sand with black spinel oxide nanoparticles is a convenient and cost-effective method for enhancing solar absorptivity while maintaining high thermal conductivity at elevated temperatures (400–700 °C)	Chung et al. [33]
Enhancing the thermal conductivity of sand by incorporating waste metallic chips sourced from machine shops	Sand mixed with scrap metal chips	Proper sand compaction can enhance thermal conductivity by up to 20 %, and that optimal bed configuration can minimize energy losses and increase storage efficiency	Tetteh et al. [20]
Verification of the suitability of different rock types and review of their thermal properties	Rock types: igneous, metamorphic, and sedimentary	Igneous and high-grade metamorphic rocks were found to be suitable for	Allen et al. [34]

Table 1 (continued)

Objectives of the study	Materials used in the study	Findings	Reference
and degradation at around 500 °C for rock-bed thermal storage in solar power plants		thermal storage applications	
Development of a new PCM-filled thermocline tank model to assess storage behavior and ascertain the viability of the latent heat thermocline concept	Conventional rock with molten-salt	The use of single PCM does not provide significant enhancement in annual storage capacity compared to conventional rocks as packing elements for an equal-sized system	Flueckiger and Garimella [35]
[36]Evaluation of sand bed heating using electric heaters with a multi-criteria heating strategy	Silica sand	The use of sand storage heating schedules will work best with RES systems (intermittent sources) or variable tariffs, while the appropriate distribution of heaters must be provided to achieve high operating efficiency and homogeneous temperature distribution	This study

200–450 °C range. These findings suggest that this method could provide a sustainable and cost-effective solution to improving energy efficiency in foundation industries.

Utilizing sand as a heat storage medium has significant potential. This technology is relatively simple but could be an efficient solution that allows the long-term accumulation of energy in the form of heat. Furthermore, sand in the TES can reach temperatures of up to 800 °C [22], and thus, the storage can also be used to supply high-temperature heat or steam for a wide range of applications, including heating buildings and industrial processes [23,24]. On the other hand, sand is not the only solid material being studied. For example, Zhou et al. [25] investigated three different materials for potential use in heat storage. These materials were sintered ore particles, aluminum oxide balls, and crushed rock. The study aimed to compare the storages' thermal properties and energy storage efficiency with these types of beds. This was done by measuring the airflow resistance and the loading and unloading capabilities of the storage tanks filled with these materials. The materials used differed in their thermal capacity, with crushed rock having the lowest heat capacity of 2727.8 kJ/(m³K) and aluminum oxide balls having the highest heat capacity of 3581.1 kJ/(m³K). Similarly, aluminum oxide balls' thermal conductivity was highest at 30 W/(mK). Each material had different parameters and could be used as required. Sintered ore particles had the highest temperature storage and thermal stability, crushed rock had the highest heat recovery efficiency (73.7 %), while aluminium oxide balls had the highest heat capacity. Schlupf et al. [26] examined the impact of fine-grained storage materials. A high-temperature packed-bed system utilizing small-grained materials, such as silica sand, quartz gravel, or basalt, to store thermal energy was analyzed. The findings indicated that materials with grain sizes of 2 mm or smaller exhibit a highly sharp temperature profile during the charging process.

Energy storage systems can be analyzed through both experimental and numerical approaches. Experimental methods involve practical testing and observation, providing direct insights into the system's

performance and behavior under various conditions. In contrast, numerical methods utilize mathematical modeling and computer simulations to predict system responses, optimize designs, and explore scenarios that may be challenging to replicate in real-life experiments for safety and efficiency reasons [26]. By integrating these two approaches, researchers can gain a deeper understanding of energy storage systems and enhance their efficiency and reliability. In addition to experimental work, several numerical studies have also been carried out. For example, Bozorg et al. [27] presented computational fluid dynamics (CFD) modelling of a TES system using liquid sodium and magnesite bricks. The geometric and flow effects analysis revealed a 95 % reduction in sodium volume while maintaining an energy recovery efficiency exceeding 99.5 % across multiple cycles. The model demonstrates minimal pressure drop and highly effective heat transfer, reinforcing the viability of this technology for high-temperature applications. Kiwan and Soud [28] shown CFD modelling of a heat storage system for beam-down solar concentrators using a mixture of sand and basalt as the storage medium. The finite element method (FEM) was used to analyze the heat conduction and to optimize the number of conductive rods and the ratio of sand to basalt. The results show that increasing the number of rods and the proportion of basalt improves the temperature distribution and increases the energy efficiency to 73 %. The CFD model confirms the effectiveness of this configuration in energy storage for CSP systems. Whereas in Ref. [29] the author developed a CFD model that considers heat conduction, convection, and energy loss and verified it experimentally. The analysis revealed that key parameters, such as sand packing density and thermal conduction coefficient, significantly impact energy storage efficiency. The modelling optimized the airflow conditions, which increased the system's efficiency by 15 %. The results confirmed that numerical simulations can effectively predict the performance of sand thermal storage under real operating conditions. The authors in Ref. [20] carried out a numerical simulation evaluation of the thermal conductivity correlations in an experimental sand TES. The CFD model incorporated heat transfer in the sand bed, the impact of grain packing on conductivity, and thermal losses within the system. It has been demonstrated that proper sand compaction can enhance thermal conductivity by up to 20 % and that optimal bed configuration can minimize energy losses and increase storage efficiency. Yousef et al. [14] analyzed the performance of a TES system using CFD modelling and response surface methodology optimization. The authors evaluated the effect of operating parameters, including input temperature, heating fluid flow rate, and the number of heat exchanger coils, on energy storage efficiency. The results showed that optimizing these parameters can increase heat recovery efficiency by 16 % and that making appropriate adjustments to the system design can significantly enhance the thermal capacity and efficiency of the system. Tuttle et al. [30] developed an efficient and accurate 1D $\times$ 1D model for simulating the transient behaviour of a packed-bed TES system. The validated model was then applied to predict the performance of various materials suitable for large-scale applications, including stone, concrete, and steel. Additionally, the impact of the uniform temperature approximation was assessed for different Biot number configurations. The proposed modelling approach achieved computation times comparable to models that do not account for intra-particle conduction, offering an 87 % improvement over previous studies on similar methods while maintaining prediction accuracy.

A brief summary of the various studies carried out on the testing and development of high temperature thermal energy storage systems is presented in Table 1.

In practical applications, the proposed sand-based storage system is well-suited for integration with thermal energy sources characterized by time variability or intermittency. From a technical standpoint, its high permissible operating temperature and thermal inertia make it compatible with renewable energy systems where the energy supply is not constant, such as photovoltaic (PV) installations coupled with electric heaters, or solar thermal collectors operating under fluctuating

solar input. In such setups, the storage unit may act as a thermal buffer, stabilizing output and enabling energy dispatch outside of generation peaks.

Additionally, the system architecture allows for effective coupling with high-temperature industrial waste heat streams, particularly those that are periodic or pulsed in nature. The embedded air channel and hybrid heating configuration facilitate both direct (hot air injection) and indirect (electrically assisted) heat input, enabling flexible thermal interfacing. These features are particularly relevant in industries such as metal processing or cement manufacturing, where high-grade residual heat is intermittently available.

Looking forward, the integration of this storage concept with dynamic heat sources will require further study of control logic, thermal regulation under fluctuating inlet conditions, and transient response optimization. Nevertheless, the thermal design and modularity of the proposed system provide a technically viable foundation for such scenarios.

Considering the current state of the art, sand as a storage medium has several advantageous properties that make it an attractive choice for thermal energy storage applications. Its high heat capacity allows for the accumulation of substantial amounts of energy within a relatively small volume, thereby enhancing its efficiency as a storage material. Additionally, sand is abundant, cost-effective, and environmentally friendly, offering a more economical alternative to other storage methods, such as water-based thermal energy storage, which operates at significantly lower temperatures. These characteristics position sand-based thermal storage systems as a viable and sustainable solution for high-temperature energy storage applications. Although a lot of work has been done so far, there are still many areas that require further attention. The main objective of the studies carried out was to develop and analyze the novel concept of charging the sand bed using electric heaters with individually controlled operating modes. A validated model was employed to determine the optimal arrangement of electric heaters, considering both the uniformity of the sand bed temperature and a multi-criteria heating strategy that incorporates the use of solar radiation and/or electricity tariffs. This solution can be considered an alternative to the currently used options of heating the storage through hot air ducts, which allow for better heat transfer inside the storage bed. On the other hand, by implementing some modifications, the proposed solution can provide hybrid electric and air heating, widening the range of its potential applications (the sand bed can be heated both by electricity generated by photovoltaic panels/electric grid and by hot air coming from heat recovery from industrial processes). A numerical model was created, and computer simulations were conducted to achieve the set objectives. Some basic simplifications have been made to allow a multivariate analysis of such thermal storage for long operating periods with relatively low computational time. The contribution of this study is at least twofold.

- the paper deals with the application of a multi-criteria heating strategy. This strategy considers solar radiation and electricity tariffs, which can be combined to achieve the most advantageous mix. In the future, it will be possible to implement artificial intelligence solutions in discharging and charging the sand bed.
- the paper presents a study of the possibility of heating sand bed using electric heaters as an alternative to the commonly used air duct systems. This solution can provide a high degree of temperature homogeneity in the sand bed, as well as precise control of the electricity used to heat the bed. In addition, the combination of electric heaters and an air duct (used here to cool the sand bed) can provide hybrid electric and air heating, widening the range of potential applications of the proposed storage system.

2. Materials and methods

The research aims to demonstrate the feasibility of thermal heat

Table 2
Analyses conducted as part of the presented study.

Analysis	Section	Description
Model validation	3.	A comparison of numerical simulation results and experimental data was conducted. The model was further verified by a mesh sensitivity analysis.
Influence of heater distribution in sand bed	5.	A variant analysis was conducted for seven different arrangements of six heaters in a sand bed.
Ability to charge the storage with different heating schedules	6.	A variant analysis was conducted to compare continuous heating, taking into account the variability of electricity purchase prices and cooperation with a photovoltaic installation.
Discharge of thermal storage	7.	An investigation was conducted into the effect of different air velocities in the heat exchanger on the operation of the heat storage system.

storage using quartz sand as a long-term, high-temperature storage medium. To perform the research, a physical laboratory prototype was developed and used to validate the numerical model. Analyses were performed using numerical models created in the ANSYS Workbench 2024 R2 (Wb) environment. The work was divided into individual stages, consisting of parametric analyses to select the best heat storage option shown in Table 2. The study included transient simulations of the storage operation (charging and discharging) for extended periods, up to three weeks. For this reason, the prepared numerical model should enable the simulations to be performed quickly with a low computational cost. To make this possible, several simplifications to the ideal variant were introduced.

2.1. Physical laboratory prototype

The physical laboratory prototype was developed using a steel cylinder-shaped tank with a volume of 205 dm³. The tank was pre-insulated with mineral wool, 10 cm thick. The tank was filled with quartz sand, with a total weight of 300 kg. Five electric heaters with individual power of 1400 W were placed within the tank to heat the sand bed. The experimental rig is illustrated in Fig. 1, and the main components of the test rig are outlined in Table 3.

The operation of the system was monitored using a dedicated control and measurement system with a modular programmable logic controller (PLC) equipped with PTTK-type temperature sensors devoted to

determining the temperature distribution within the bed. The operation of the heaters was controlled by relays which switched them on and off according to an assumed time schedule (set individually for each series of measurements).

Considering power of the electric heaters and time of their operation, it was possible to determine the amount of input energy delivered for charging the sand bed. Ongoing observation of operating parameters and data archiving with 5s intervals was possible thanks to a special computer visualization prepared in the CoDeSys environment.

2.2. Implementation of sand bed parameters into the model

A sand bed should be considered as a porous body. This means that its thermodynamic properties are closely dependent on the size and type of sand grains. The smaller the grain size of the bed, the less air there is between the grains, which results in a higher effective heat transfer coefficient and specific heat of the entire bed. Three different ways of modeling a sand bed have been identified. The first, which assumes knowledge of the sand particle size and its exact parameters, is based on the discrete modeling of particles suspended in a fluid [37]. This approach, however, is computationally very demanding and is not suitable for variational analysis based on transient simulations. The second approach is based on modeling the bed as a porous material (porous zone) [38]. This means accounting numerically for the pressure drop flowing through the air pores. This drop is calculated according to Darcy’s law [39], which requires knowledge of the porosity and permeability of the sand bed. In the first attempts to carry out simulations for this model, it was observed that the complexity of airflow phenomena, including the convection process, is also inadequate for analysis. The simulation time would not allow for a multi-variant

Table 3
The main elements of the test rig used during the tests.

Element	Description
Heat storage	Cylindrical metal container: • Diameter of 56 cm • Height of 70 cm • Insulation 10 cm of mineral wool, $\lambda = 0.040$ W/(mK)
Electric heaters	Electrical power = 5 x 1400 W
Sand	Mass of sand = 300 kg
Temperature sensors	K-type thermocouple sensors (NiCr-NiAl) with a measuring range from -200 °C to 1150 °C and tolerance ± 2.0 °C or $\pm 0.0075 \times [t]$, where t is the measured temperature in °C

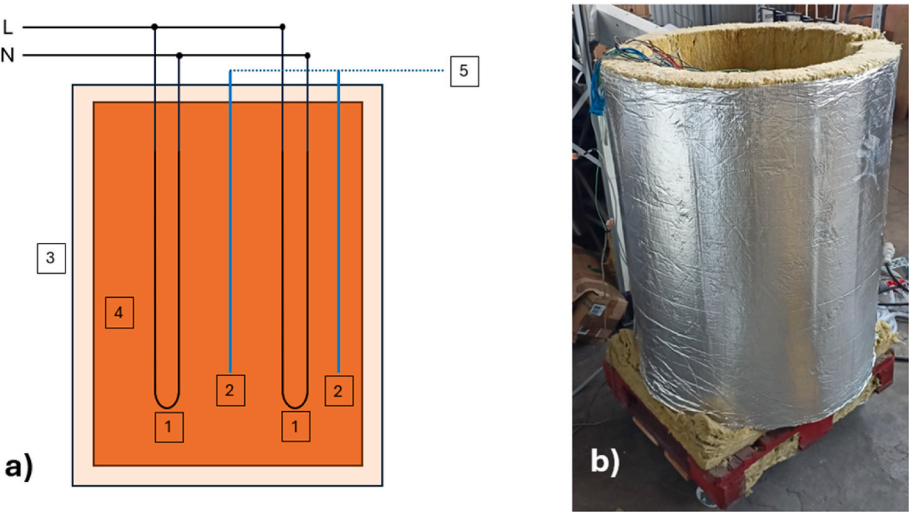


Fig. 1. View of the experimental rig - schematic cross-section (a) and real view (b): 1 - electric heaters, 2 - temperature sensors, 3 - tank with insulation, 4 - sand, 5 - automatic data logger system based on PLC.

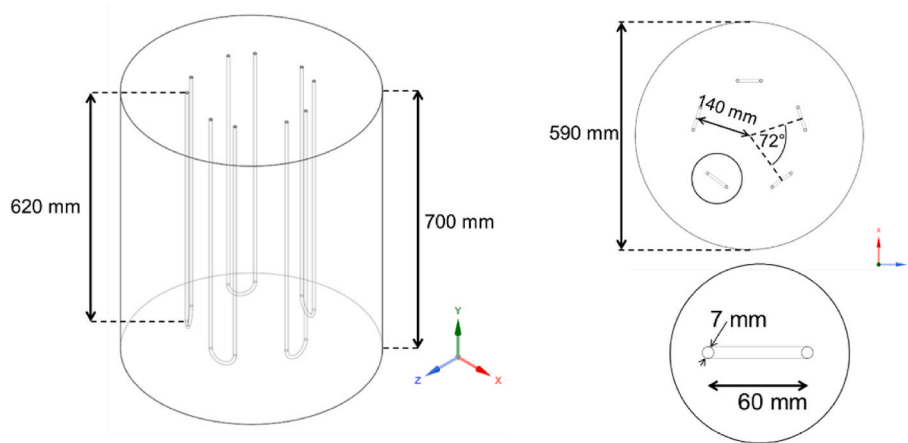


Fig. 2. Geometry of the numerical model, including dimensions of individual elements.

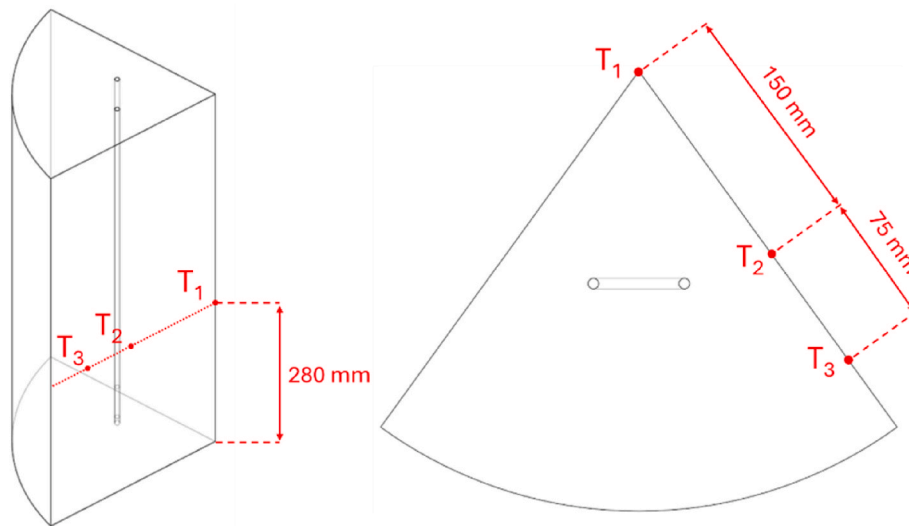


Fig. 3. Geometry of one-fifth of the volume of the sand bed used in the calculations along with the location of the measurement points.

analysis. The last approach is based on a significant simplification of the properties of the sand bed. It is possible to determine the substitute thermodynamic parameters of the bed. This requires separate studies of the bed for a given gradation and sand type, so it is not a universal method. Baumann and Zunft [40] demonstrated the use of the “hot wire” method when studying the thermal conductivity of a sand bed. This approach allows the storage material to be modeled as an all-solid with substitute thermodynamic parameters. For the model used in the following article, the last method of implementation of sand parameters was chosen. Data obtained by Baumann and Zunft show that the coefficient of thermal conductivity varies parabolically or approximately linearly over the temperature range of 300–1073 K. This is an important observation because when quartz sand is heated, a polymorphic phase transformation phenomenon occurs. At a temperature of about 846 K, alpha quartz transforms into beta quartz, changing its structure and physical and thermal properties. This can be seen in the graph of the change in specific heat, where at $T = 846$ K a jump in value is apparent, and then for $T > 846$ K the resulting specific heat of sand is lower than for $T < 846$ K. During the development of the model, it was decided to simplify this relationship and follow the function present for specific heat for temperatures ranged from 323 to 846 K. The operating conditions of the modeled sand bed for most of the time include temperature values in this range by which this simplification is possible.

A linear trend line was assigned for specific heat (Eq. (1)) and

thermal conductivity (Eq. (2)), which allowed them to be imported into the numerical model.

$$C_p(T) = T + 427 \left[\frac{J}{kgK} \right] \quad (1)$$

$$k_c(T) = 0.0006T - 0.0102 \left[\frac{W}{mK} \right] \quad (2)$$

2.3. Numerical model

a) Geometry

The geometry of the model was created using the native Wb program, SpaceClaim. It contained only the domain of the sand bed - a solid. The insulation layer of the storage was assumed to be an infinitely thin wall to limit the number of mesh cells. The geometry along with its dimensions is presented in Fig. 2.

The heating elements were implemented as a cutout of their volume from the sand bed. Their location coincided with the validation experiment (see Figs. 1 and 3). The structure of the investigated heat storage is axially symmetrical. It is possible to divide it into five equal elements. This made it possible to simplify the entire model and perform calculations for only one-fifth of the entire volume. The section of the storage

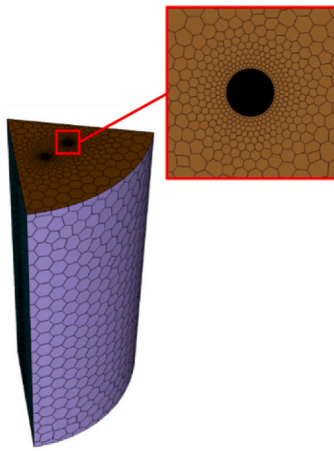


Fig. 4. Numerical mesh created for validation purposes.

Table 4
Summary of the parameters of the numerical mesh used for validation.

Parameter	Value	Allowed value according to ANSYS Workbench Manual documentation [36]
Number of nodes	4 279 874	–
Number of cells	828 081	–
Max/Min orthogonal quality	0.997/0.5	>0.05/~1
Avg orthogonal quality	0.960	>0.6
Min/Max skewness	0.00151/0.500	~0/<0.95
Avg skewness	0.0397	<0.4

Table 5
A summary of the boundary conditions introduced into the model with a description.

Boundary	Boundary condition	Definition	Description
Heater surface	Wall, fixed heat flux	$Q = 4.77 \cdot 10^4 \left[\frac{W}{m^2} \right]$	Electrical power of heater (1400 W) distributed on heater surface ($2.93 \times 10^{-2} m^2$)
Dividing wall	Symmetry	[37]	Enabling the correct temperature distribution throughout the storage volume
Upper surface of sand bed	Wall, convective heat transfer	$Nu = \begin{cases} 0.54 \cdot Ra^{1/4} & \text{for } 10^4 < Ra < 10^7 \\ 0.15 \cdot Ra^{1/3} & \text{for } 10^7 < Ra < 10^{11} \end{cases}$	Heat exchange by natural convection over the entire surface
Side surface of sand bed	Wall, convective heat transfer	$Nu = \left\{ 0.825 + \frac{0.387 \cdot Ra^{1/6}}{\left[1 + \left(\frac{0.492}{Pr} \right)^{9/16} \right]^{8/27}} \right\}^2$	
Bottom surface of sand bed	Wall, convective heat transfer	$Nu = 0.27 \cdot Ra^{1/4} \text{ for } 10^5 < Ra < 10^{11}$	

used for calculations and the position of the thermocouples, which coincided with the validation experiment, is shown in Fig. 3.

b) Mesh

The numerical mesh was created in ANSYS Fluent Meshing software. In addition to the global settings, the local aggregation of mesh cells near the heaters was introduced into the algorithm. A sensitivity analysis of the mesh determined the optimal maximum global cell size to be 0.05 m, along with the corresponding local aggregation, which resulted in a maximum cell size on the heating surface of 0.0005 m. The grid consisted exclusively of polyhedral cells, totaling 828 081 pieces. The finished numerical grid is shown in Fig. 4, and selected key parameters of the grid are summarized in Table 4.

c) Mathematical model

Due to the simplifications described in Section 2.2, the entire computational domain consisted of a solid. ANSYS Fluent software, which is typically used for CFD analyses, was used for the calculations. In the model used for validation, no flow is observed in the computational domain. However, this software was chosen because it allows for fast heat transport calculations in a solid and, in addition, in further analyses there will be a heat exchanger in the model, where air flow will be present. By using this software for all analyses, it was possible to ensure adequate convergence between their results and reality.

The operation of the ANSYS Fluent solver is based on the basic governing equations [41]: continuity and conservation of momentum and energy. When analyzing only heat transport in a solid, the solver uses only the latter represented by the heat conduction equation (Eq. (3)). It is worth noting that this form of the equation does not assume terms related to fluid flow.

$$\rho \frac{\partial T}{\partial t} = \frac{k_c}{C_p} \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right) \quad (3)$$

The outer walls of the model (insulated surfaces and electric heaters) were implemented into the model as infinitely thin surfaces and described by the corresponding boundary condition (see Table 5). The locations of all boundary conditions within the computational domain are shown in Fig. 5. The heat storage in the experiment was located in a large room with a constant temperature thus main heat transfer process on its walls was the natural convection phenomenon. To properly implement this mechanism into the model, it is necessary to determine the heat transfer coefficient h_c [W/(m²K)]. This is possible using the Nusselt correlation according to the formula [42]:

$$h_c = \frac{Nu \cdot \lambda}{L} \left[\frac{W}{m^2 K} \right] \quad (4)$$

Table 5 summarizes the formulas to determine Nu for the horizontal and vertical walls of the thermal storage [42].

Numerical simulation made it possible to obtain the temperature distribution inside the sand bed. Particularly important were the temperature values at the measurement points consistent with the experiment (T_1, T_2, T_3 [K]), the maximum and average temperature of the heaters ($T_{h,max}, T_{h,ave}$ [K]) and the maximum and average temperature of the sand ($T_{s,max}, T_{s,ave}$ [K]). This data allowed to conduct mesh sensitivity analysis, as well as validate the numerical model.

d) Results of mesh sensitivity analysis

A sensitivity analysis was conducted to determine the optimal cell size for the numerical grid. The process was divided into two stages: adjusting the global cell size and modifying the local aggregation size in the heater region (see Fig. 6).

The first stage of the analysis showed that changing the global cell

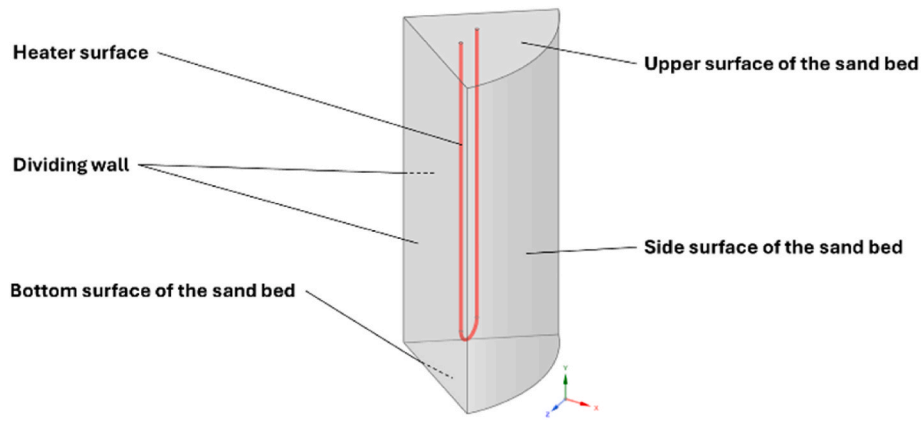


Fig. 5. Location of individual boundary conditions on the geometry of the computational domain.

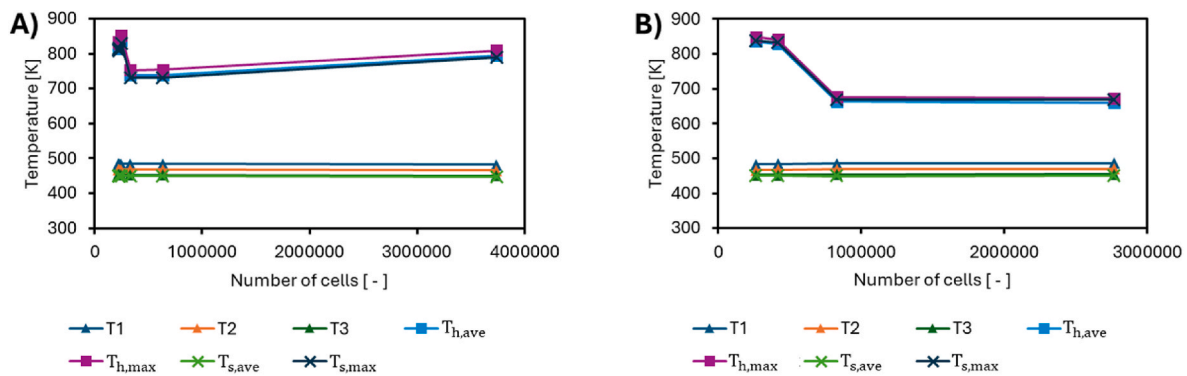


Fig. 6. Comparison of temperature values at measuring points depending on the number of cells of the numerical mesh. A) modification of global cell size, B) modification of local cell size on heater surface.

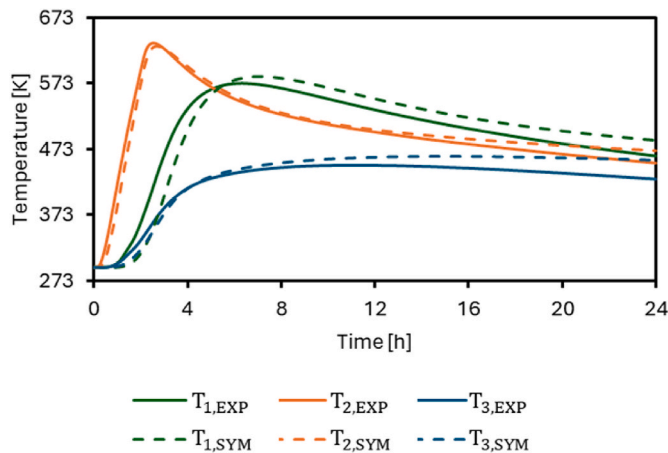


Fig. 7. Comparison of temperature readings at measurement points from experiment and simulation.

size in the range of 0.1–0.0025 m has no direct effect on the values of T_1 , T_2 , T_3 and $T_{s,ave}$. The temperature readings at the heater surface $T_{h,max}$, $T_{h,min}$ and the maximum sand temperature $T_{s,max}$, which is close to the heaters, do not stabilize with increasing global cell size. For this reason, a second analysis of the effect of local grid density was decided. It revealed that the rest of the parameters show stabilization for local cell sizes smaller than 0.0005 m. The final mesh settings that were implemented into the model assumed a global cell size of 0.05 m and a local sizing of 0.0005 m on heater surface

3. Numerical model validation

Data collected from experiment were used for the validation of the simulation results. Temperature readings $T_{1,SYM}$, $T_{2,SYM}$ and $T_{3,SYM}$ from the simulation were compared with values from the experiment ($T_{1,EXP}$, $T_{2,EXP}$ and $T_{3,EXP}$) at the same location in the bed (see Figs. 3 and 7). Both the measurement for the experiment and the simulation duration involved 2 h of bed heating and 22 h of temperature stabilization.

The graph shows some differences in the values obtained from the validation experiment and the simulation. The average temperature differences for measurements at T_1 , T_2 and T_3 were, respectively: 14 K, 9 K and 20 K. For each measurement point, a certain discrepancy is apparent in the first hours of simulation, which decreases until the maximum temperature is reached at that point and then increases again as the bed cools down. The thermodynamic parameters of the sand bed are from Ref. [40] Their implementation into the model gave a very high convergence, but it would be misleading to say that the quartz sand used by Bauman et al. has identical properties to the one used in our experiment. The deviations observed in the graph can be directly related to the slightly different value of the bed's thermal diffusivity between the paper and the experiment. Most likely, the value obtained by Bauman et al. was lower. This agrees with the indications from the graph (see Fig. 7). At the beginning of the analyzed period in the simulation, the sand heats up more slowly. Similarly, it cools more slowly after the heaters are turned off. This may indicate poorer thermal penetration properties of the sand for Bauman's data. However, these differences do not significantly impact the presented studies. It focuses mainly on variational analysis. Maintaining high convergence and the nature of the temperature change over time is sufficient to observe the relationships presented in the paper. The studies presented in this work are mainly

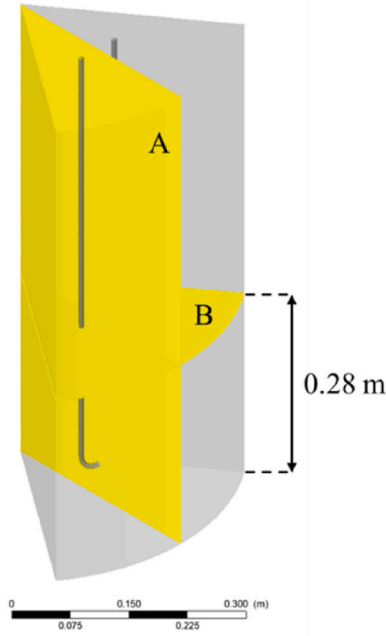


Fig. 8. Location of cross sections from which temperature distributions were taken. A – vertical plane, B – horizontal plane.

aimed at displaying the basic geometric parameters affecting the operation of the heat storage. These relationships can be observed with the help of variational analysis, which is possible using the numerical model validated in this way. In the future, more advanced studies are planned that take into account the implementation of the obtained results to a real setup. Particular attention should then be paid to the appropriate measurement of the thermodynamic parameters of the bed.

The simulation made it possible to obtain the temperature distributions shown in Fig. 9. Contours were taken at two cross-sections of the storage geometry, as shown in Fig. 8.

Analyzing the data presented in Fig. 9, it can be observed that at $t = 2$ h, immediately after the heaters are turned off, a significant temperature difference is visible in the bed volume. After 10 h of stabilization, an even temperature distribution is observed on the horizontal section, with the largest reading in the center of the bed and the smallest on the walls of the container. Such a distribution persists until the end of the simulation, accompanied by a decrease in the average temperature of the bed, as well as in the temperature difference between the coldest and warmest areas of the storage. The temperature distribution in the vertical section of the container is analogous. It is possible to isolate the hot core of the sand bed and the decrease in its temperature as it approaches the walls of the storage. The temperature values obtained at various simulation moments are presented in Table 6.

The results of this analysis reveal a high potential for heat storage in sand. Despite suboptimal insulation layer thicknesses, it is possible to maintain high sand temperatures for an extended period. This is made possible by the low thermal conductivity coefficient of the sand, which forms a natural insulation barrier near the walls of the storage, thereby increasing the time required to cool the bed. It is recognized that the thermophysical properties of the sand bed used in the numerical model were derived from secondary sources [40] rather than direct experimental measurements of the actual material. While validation against experimental temperature profiles confirmed the model's reliability for the current study's scope, future work will involve direct characterization of the used sand, including temperature-dependent measurements of thermal conductivity and specific heat capacity. This will further enhance the model's material fidelity and predictive accuracy. The validated model described above was used to conduct a variant analysis of the sand heat storage operation. Different operation variants were

implemented, which are described in the following sections.

4. Effective thermal conductivity coefficient of the enclosed fluid

To extend the previously developed model, a pipe heat exchanger was added to the bed to allow for subsequent heat removal from the storage. For this purpose, air was chosen as the working medium, which, unlike other low-cost heating media, can be heated to the operating temperatures predicted for the final heat storage (~ 850 K). Simulations performed as part of the variant analysis assumed large time intervals (up to 168 h). For this reason, it was crucial to implement a simplification that allowed omitting the direct calculation of convection processes inside the new fluid domain.

Despite the enclosure of the air portion inside the piping, when heating the storage, the air moves due to the natural convection process. However, it is possible to account for it with the help of the effective heat transfer coefficient k_{eff} [W/mK], which can be determined according to the equation [42]:

$$k_{eff} = k_c \cdot Nu \quad (5)$$

The Nusselt number directly indicates whether the heat transfer inside the fluid occurs inclusively by conduction ($Nu = 1$) or if a convection process is also observed ($Nu > 1$). It is directly possible to determine Nu as:

$$Nu = \frac{h_c \cdot L}{k_c} \quad (6)$$

Unfortunately, the heat transfer coefficient itself is not a material value and is closely related to the nature of the flow. It can be determined with the help of the Nusselt correlation [42], which consequently allows to determine the corresponding expression for Nu . Due to the great complexity of the convection phenomenon, the different cases of its occurrence are characterized by distinct approaches in estimating the magnitude of h . The considered case in the presented model can be compared to the approach in calculating the effective conduction coefficient in a fluid enclosed in a container inclined at a certain angle. In this case, it is possible to determine Nu as:

$$Nu = 1 + 1.44 \left[1 - \frac{1708}{Ra \cdot \cos \theta} \right]^+ \left(1 - \frac{1708 (\sin 1.8\theta)^{1.6}}{Ra \cdot \cos \theta} \right) + \left[\frac{(Ra \cdot \cos \theta)^{\frac{1}{3}}}{18} - 1 \right]^+ \quad (7)$$

Gathering the individual expressions into a bracket with a “+” symbol means that if the expressions in these brackets take a negative value, they should be set to zero.

The Rayleigh number (Ra) is determined as the product of Grashoff's (Gr) number and Prandtl's (Pr) number, and when expanded, the expression for Ra takes the form:

$$Ra = \underbrace{\left\{ \frac{g \beta \Delta T L^3}{\nu^2} \right\}}_{Gr} \cdot \underbrace{\left\{ \frac{\nu}{\alpha} \right\}}_{Pr} \quad (8)$$

where: ΔT – temperature difference between two sides of enclosure [K].

Properly introduced effective heat transfer coefficients should be calculated for each fluid cell. Unfortunately, the problem lies in accurately determining ΔT for the case under consideration due to the large temperature distribution of the sand bed. For this reason, the effect of implementing the effective conduction coefficient on the obtained results was investigated. This analysis was conducted for the first variant of heater distribution, as presented in Section 5. For this purpose, two simulations were compared: the first assumes leaving the air coefficient as a table value, and the second considers the highest predicted value of the conduction coefficient. It is worth noting that in the considered

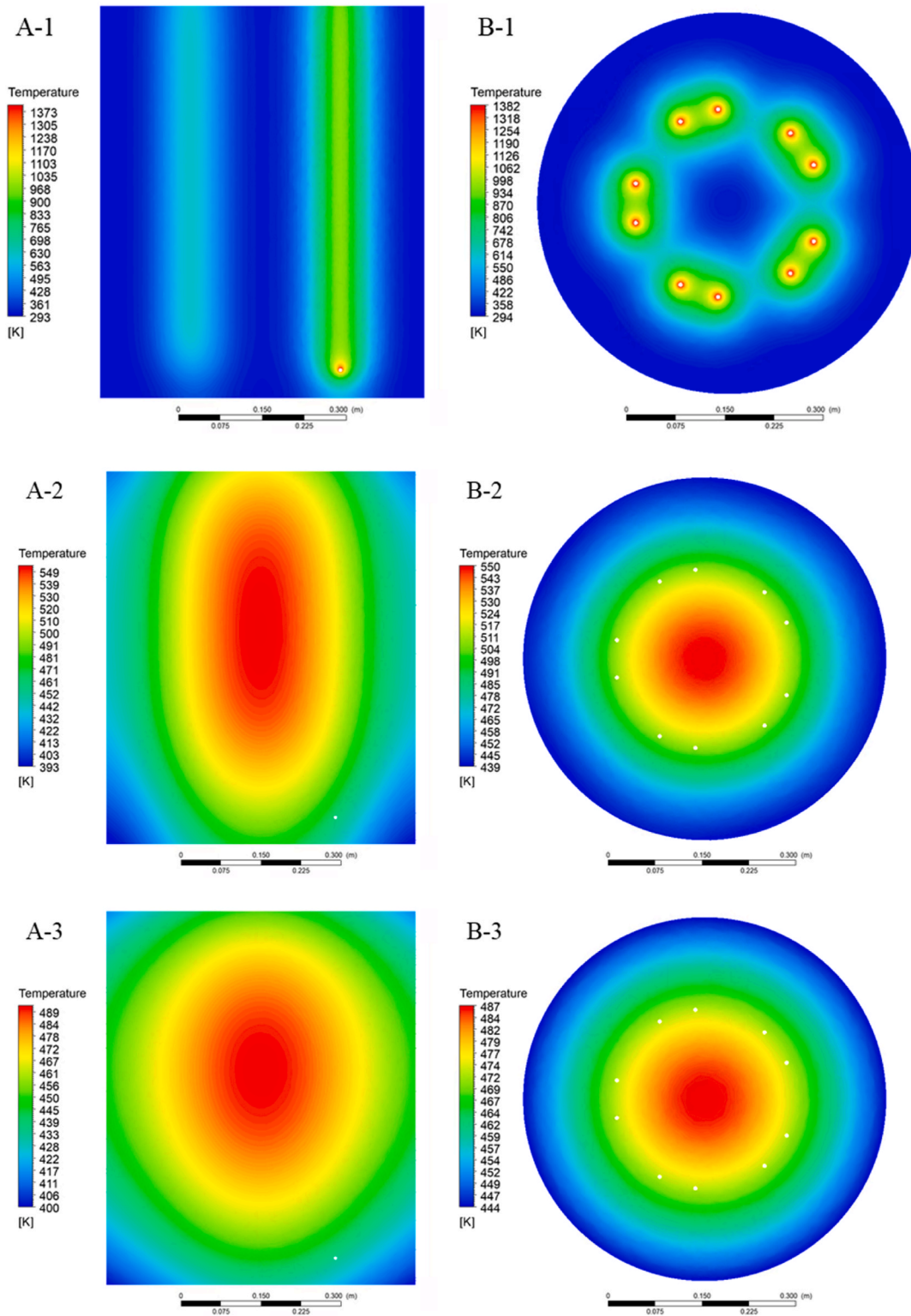


Fig. 9. Temperature distribution inside the sand bed for cross sections A and B. A-1 and B-1 – time of simulation $t = 2$ h, A-2 and B-2 – time of simulation $t = 12$ h, A-3 and B-3 – time of simulation $t = 24$ h.

simulation, the air parameters were assumed to be constant (including specific heat and density). For this reason, the effective conduction coefficient depends only on ΔT . The graph below (see Fig. 10) shows the dependence of k_{eff} on ΔT .

The critical maximum value of the conduction coefficient, enclosed air will reach for the maximum temperature difference. For the model under consideration, this is the difference between the initial tempera-

ture of the simulation $T_0 = 293$ K and the predicted final temperature of the sand bed $T_{s,end} \approx 407$ K, that is, $\Delta T \approx 114$ K. For this value, $k_{eff} \approx 0.21 \frac{W}{mK}$. The simulation results are shown in Fig. 11, along with the calculated relative error between the models for each value of thermal conductivity.

The simulation results show a very small effect of changing the

Table 6

Comparison of temperature values inside the sand bed for different time steps.

Time [h]	Average sand temperature [K]	Maximum sand temperature [K]	Minimum sand temperature [K]	Temperature difference inside sand bed [K]
2	470	1392	293	1099
12	469	554	391	163
24	436	491	400	91

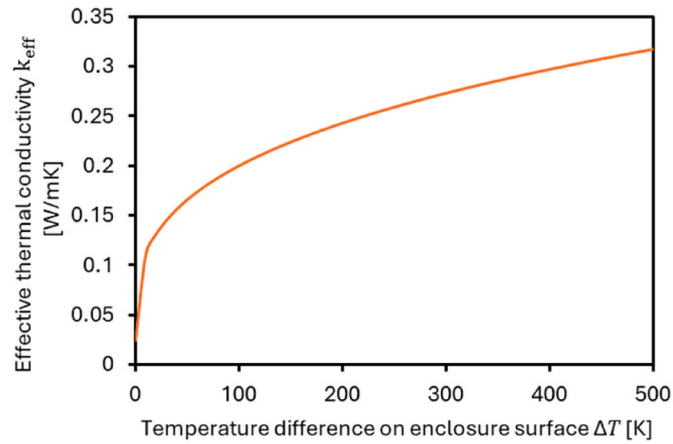


Fig. 10. Dependence of the effective heat transfer coefficient on the temperature difference on opposite sides of the enclosure.

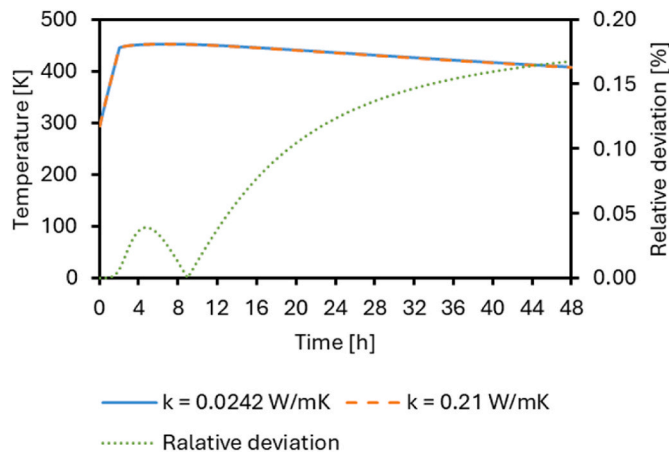


Fig. 11. Comparison of simulation results for variants assuming different fluid heat transfer coefficients, with relative deviation highlighted.

thermal conductivity of the air, with the largest relative deviation not exceeding 0.2 %. This allows to conclude that it is possible to carry out a correct simulation of the model in which a tabular value of thermal conductivity characterizes the air. It is also worth mentioning here that during the heating process, the temperature difference between the sand and the fluid decreases over time. This means that the more homogeneous the temperature distribution inside the sand bed, the smaller the contribution of natural convection to heat transport inside the closed fluid domain.

5. Effect of heater distribution on storage charging efficiency

The first variant analysis of the prepared numerical model focused on determining the optimal distribution of electric heaters within the sand bed. The geometry of the storage was modified for future analyses by the

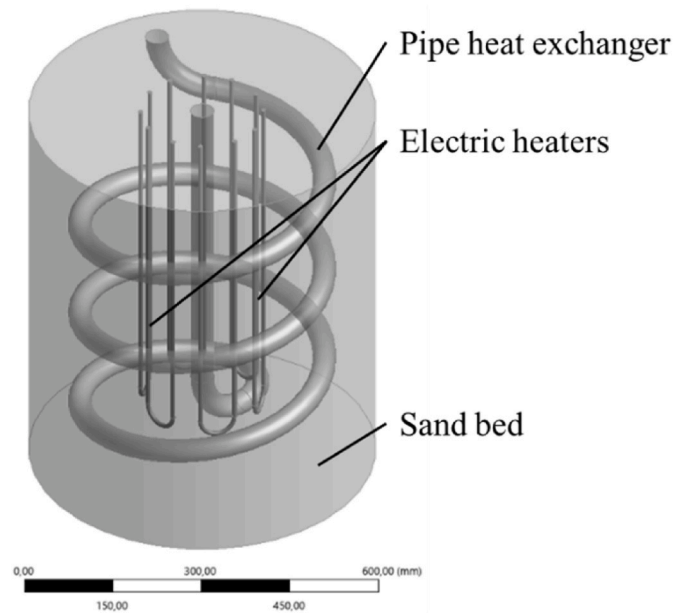


Fig. 12. Modified geometry of the model with pipe heat exchanger.

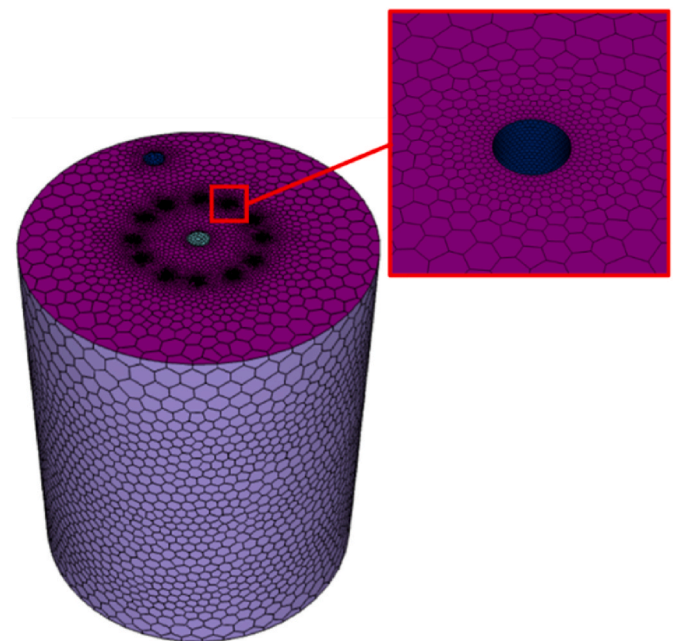


Fig. 13. Numerical mesh of modified model with pipe heat exchanger.

Table 7

Mesh parameters of modified model with pipe heat exchanger.

Parameter	Value	Allowed value according to ANSYS Workbench Manual documentation [36]
Number of nodes	25 334 927	–
Number of cells	4 903 347	–
Max/Min orthogonal quality	0.998/0.5	>0.05/~1
Avg orthogonal quality	0.960	>0.6
Min/Max skewness	0.00134/0.498	~0/<0.95
Avg skewness	0.039	<0.4

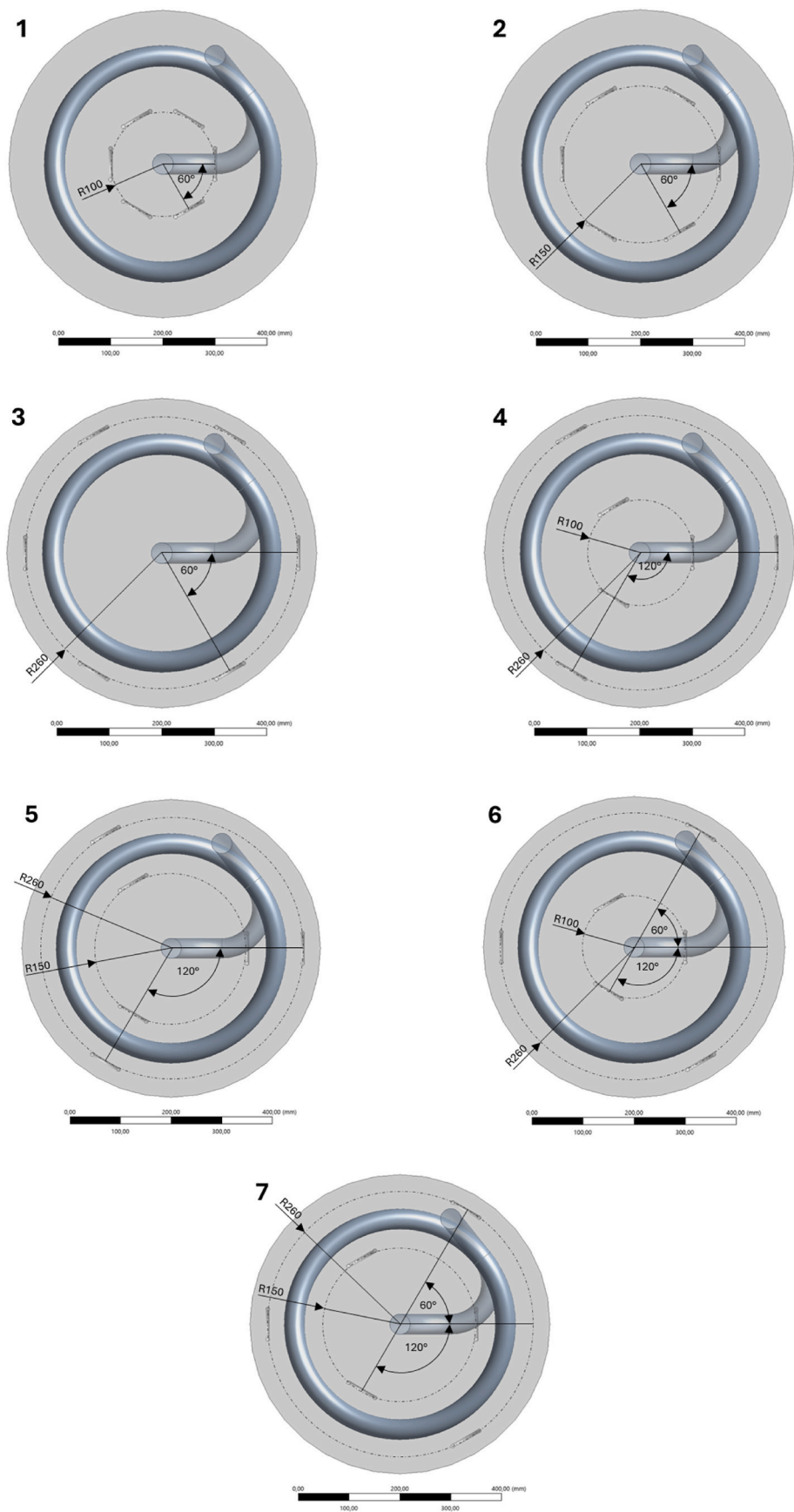


Fig. 14. Geometries of different variants (1–7) of heater distribution inside the sand bed.

Table 8

Results of variant analysis of the distribution of heaters inside the sand bed.

Variant	Storage ratio			ΔT [K]			$T_{h,max}$ [K]
	t = 2h	t = 24h	t = 48h	t = 2h	t = 24h	t = 48h	
1	0.998	0.860	0.685	1275	400	343	1306
2	0.998	0.841	0.669	1207	382	341	1240
3	0.986	0.768	0.608	1315	350	334	1348
4	0.992	0.815	0.647	1260	377	339	1291
5	0.992	0.805	0.639	1263	367	338	1297
6	0.992	0.815	0.647	1316	376	339	1348
7	0.992	0.805	0.639	1316	364	337	1348

addition of a pipe heat exchanger. The dimensions of the geometric model (tank size and heater size) have remained unchanged (see Fig. 12).

In addition to the implementation of the air exchanger, the number of electric heaters and their respective unit power were modified. Six heating elements with a specific power of 1000 W were introduced into the sand bed. This procedure was intended to allow a simpler reference of the obtained results to the initial variant, as well as to facilitate an even distribution of the heaters in the bed. The change in geometry automatically contributed to a change in the calculation mesh, which was performed in accordance with the previous sensitivity analyses. It is worth mentioning that due to not considering convective phenomena inside the fluid domain, the air was stationary during the simulation. For this reason, the size of the fluid cells equal to the size of the solid cells chosen based on the previous sensitivity analysis was sufficient. The computational grid is shown in Fig. 13 and its parameters in Table 7. It is worth noting that the presence of piping within the sand bed resulted in an asymmetrical geometry. For this reason, tank had to be modeled in its entirety, relative to the validation model, in which it was only one-fifth of its volume. This fact directly affected the number of mesh cells.

Seven different variants of heater distribution inside the sand bed were investigated (see Fig. 14). The settings of the solver changed compared to the validation simulation. However, these were not significant enough to affect the reliability of the model. The symmetry boundary condition is no longer present in the model because the simulation encompasses the entire storage domain, rather than just a part of it. The air inlet and outlet were closed and assumed to be walls with parameters analogous to the rest of the walls of the computational domain.

The analysis included a two-stage simulation: heating for the first 2 h and temperature stabilization for the next 46 h. Results for each variant were compared using temperature indications: minimum, maximum and average temperature of the sand bed ($T_{s,min}$, $T_{s,max}$ and $T_{s,ave}$ [K]) and temperature difference $\Delta T = T_{s,max} - T_{s,min}$ [K]. For each simulation, the amount of net energy stored in the bed E_{nett} [MJ] was also determined. In addition, the Storage Ratio (SR) parameter was introduced to show the efficiency of the storage operation for the analyzed time. Its value was calculated according to the equation:

$$SR = \frac{E_{nett}}{\text{Energy input}} \quad (9)$$

Energy input for the analyzed case was related only to the operation of electric heaters, that is, it was a constant value and amounted to 43.2 MJ. An important fact that has not yet been mentioned is the temperature to which the electric heater can be heated. Due to the low thermal conductivity of sand, the heat from the heating element is not withdrawn as efficiently as for direct heating of fluid. An electric heater placed in the sand can theoretically reach very high temperatures for which the heating wire itself would melt. For this reason, the last important parameter dictating which of the selected variants would be best was the maximum temperature of the heater at the end of the heating time $T_{h,max}$ [K].

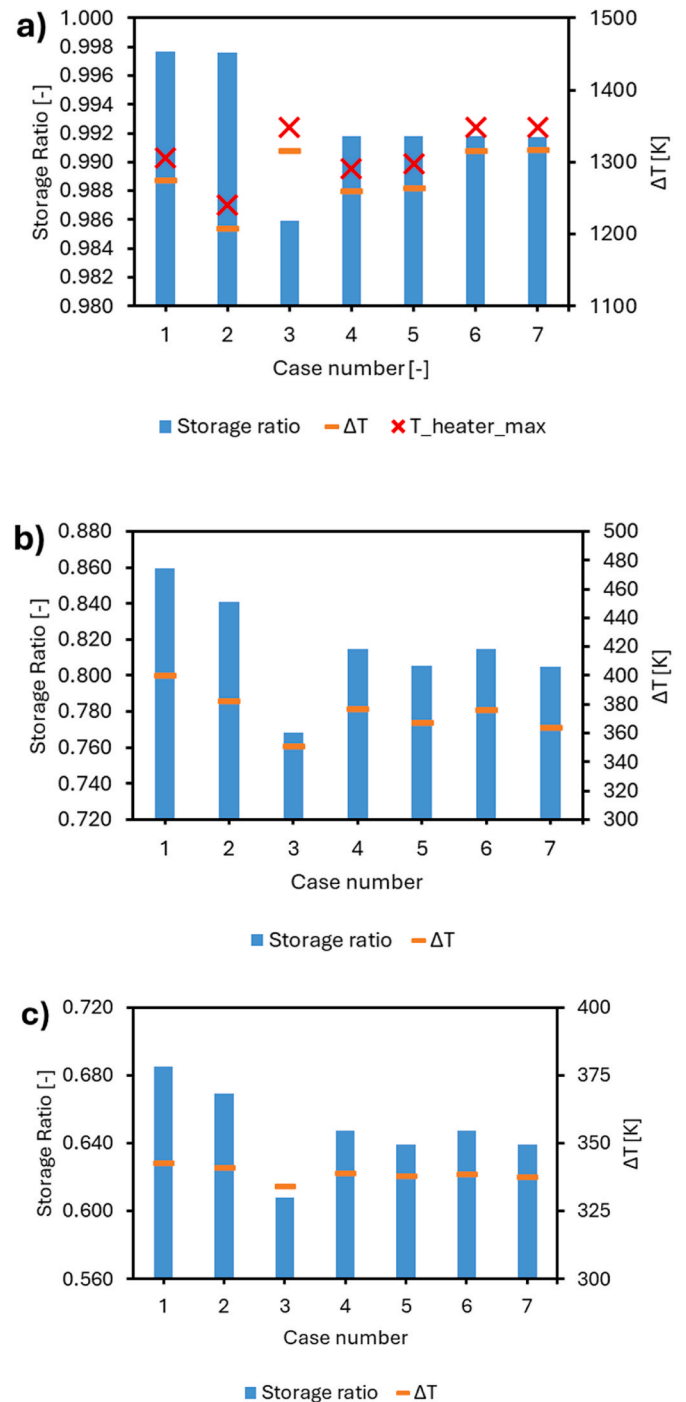


Fig. 15. Values of storage ratio, temperature distribution and maximum heater temperature for each analysis variant at: a) t = 2 h, b) t = 24 h, c) t = 48 h.

Simulation results were collected for different time steps: t = 2 h (at the end of heating), t = 24 h and t = 48 h (end of simulation). All data are summarized in Table 8. To facilitate the selection of the best variant, graphs were created showing the obtained parameters for each analyzed time step (Fig. 15).

The second variant was chosen as having the best parameters. It was characterized by one of the highest Storage Ratio values with the lowest maximum heater temperature and one of the most stable temperature distributions. Other variants showed that placing the heaters near the center of the sand bed results in lower losses to the environment during the analysis period. Unfortunately, their proximity to each other significantly increases their temperature. Despite the largest distances

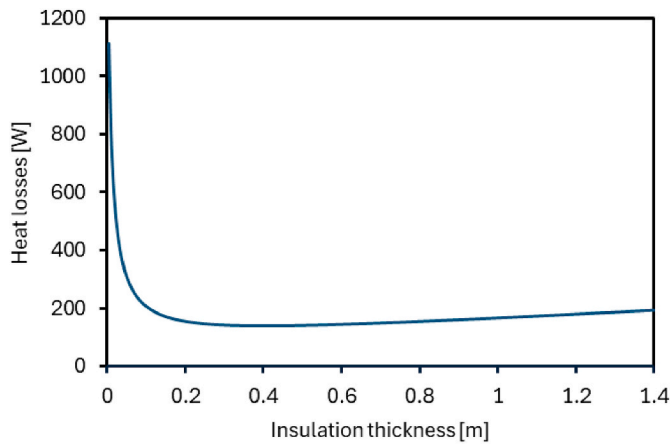


Fig. 16. Dependence of the magnitude of heat loss flux on the outer walls of a cylindrical container on the thickness of the insulation layer.

between the heaters for variant 3, both their highest temperature and the highest losses to the environment are observed. This fact is related to their placement close to the walls of the storage. Insulation does not allow even distribution of heat flux around the heaters by which they heat up more. Mixed variants (4–7) are characterized by a similar problem due to the proximity of some of the heaters to the walls of the container. SR for all analyzed variants after 48 h of simulation indicates significant heat losses to the surroundings, regardless of the heater placement. This is due to the suboptimal thickness of the insulation layer on the storage walls and should be taken into account in further analyses.

6. Analysis of the possibilities of different heating schedules

The next analysis focused on verifying the feasibility of supplying energy to a sand heat storage system under different heating schedules. To prepare a correct model for these purposes, two problems observed in previous analyses needed to be addressed. The first is the insulation layer of the storage walls. To this point, the walls in the model have been described analogously to the validation experiment. This resulted in significant energy losses to the environment and, consequently, low SR values (see Table 8). It was decided to thicken the insulation layer in future models. Due to the cylindrical shape of the storage, increasing the thickness of the insulation also increases the heat transfer surface area, thereby increasing losses to the surroundings. It is possible to determine the optimal insulation thickness for which losses reach a minimum value by solving the differential equation of thermal conductivity for a cylinder (Eq. (10)) and a flat wall in the case of bases (Eq. (11)).

$$\frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) = 0 \quad (10)$$

$$\frac{\partial^2 T}{\partial x^2} = 0 \quad (11)$$

Table 9

Comparison of different sand bed heating schedules.

Case	Heater restart temperature [K]	Storage Ratio [–]	Last time step temperature of sand bed [K]	Average number of activations per heater [–]	Stored energy		Energy density	
					[MJ]	[kWh]	[MJ/m ³]	[kWh/m ³]
Continuous heating	673	0.826	665.87	34	102.51	28.47	554.66	154.07
	773	0.828	750.64	35	128.38	35.66	694.67	192.97
	883	0.832	847.09	251	159.02	44.17	860.44	239.01
G12	673	0.829	661.85	31	101.32	28.15	548.25	152.29
PV _s		0.829	656.15	30	99.63	27.67	539.09	149.75
PV _w		0.840	648.40	27	97.32	27.03	526.62	146.28

where: r – distance from the axis of the cylinder [m], x – distance from inner side of insulation [m].

It should be noted that the equations written out above are a simplified form of the full heat conduction equation and incorporate assumptions: the specific heat, thermal conductivity and density of the material are constant values, and the analysis itself applies to a stationary problem. Despite the simplifications, the model calculated in this way enables the illustration of the problem of increasing the heat transfer surface area. The results of the analysis are shown in the graph of the dependence of the heat loss flux to the environment on the thickness of the insulation layer (see Fig. 16). The minimum of the illustrated function can be observed for a thickness of approximately 0.403 m. This value was implemented into the model for each subsequent analysis.

The second problem was that the temperature of the heaters was too high during operation. Based on the data provided by the heater manufacturers and experience after the performed tests, the maximum safe temperature that the heating element should withstand was ~ 1073 K. For this reason, the model required the implementation of appropriate control logic that would prevent the heating element from melting under real conditions. A two-state (binary) control was decided upon. The operation of the heater was related to its average temperature according to the relationship presented in Eq. (12).

$$S(T_n) = \begin{cases} 1, & \text{if } T_n \leq 673 \text{ K} \\ S(T_{n-1}) & \text{if } 673 \text{ K} < T_n < 1073 \text{ K} \\ 0 & \text{if } T_n \geq 1073 \text{ K} \end{cases} \quad (12)$$

where: $S(T_n) \in \{0, 1\}$ – state of operation of the heater for time step n [–], T_n – average temperature of the heater for time step n [K], $S(T_{n-1})$ – state of operation of the heater for the previous time step ($n - 1$) [–].

It should be noted that implementing this approach to controlling the heaters' operation is quite challenging in practice. The high temperature

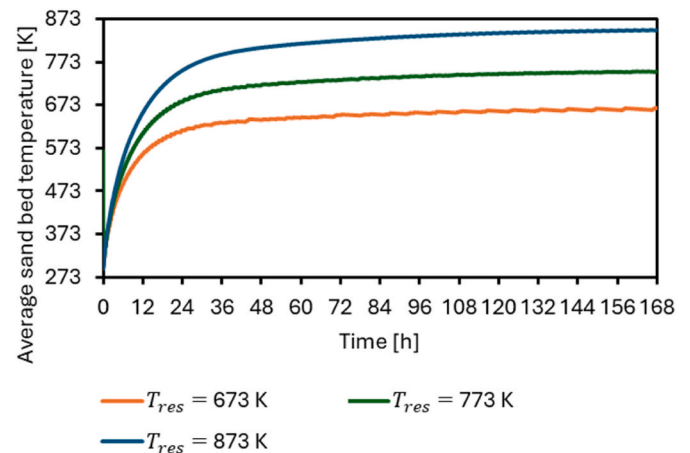


Fig. 17. Variation of average sand bed temperature for variants assuming continuous heating.

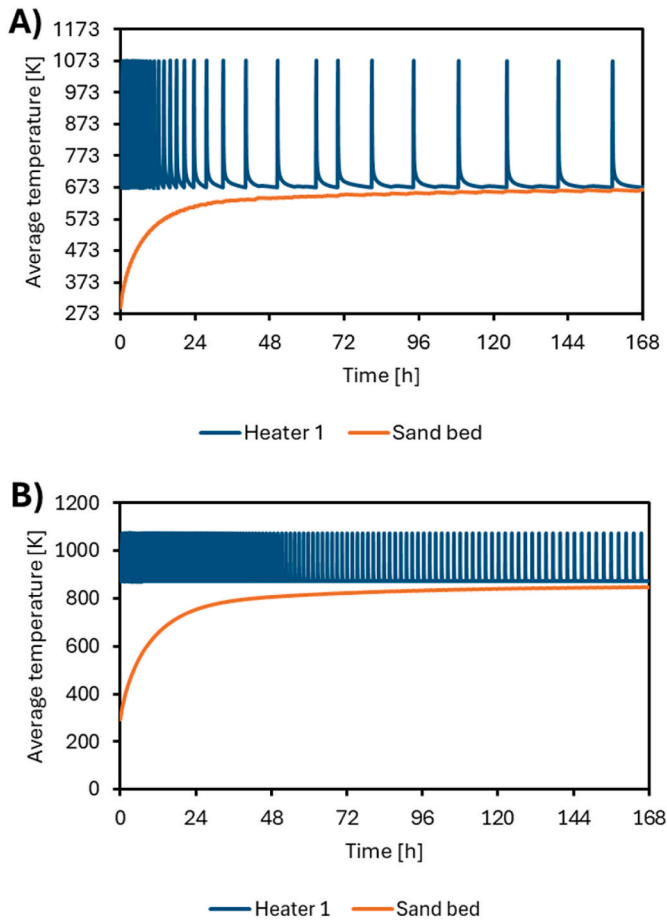


Fig. 18. Variation of the average temperature of the sand bed. A) minimum heater temperature $T_{res} = 673$ K, B) minimum heater temperature $T_{res} = 873$ K

poses a significant risk of damage to sensors placed directly on the heater. Additionally, averaging the temperature on its surface could prove impossible or subject to significant error. In the future, it is planned to measure the thermal penetration capability of the sand bed and locate the sensors at a known distance from the heating element. An accurate simulation of the operation of the storage will make it possible to relate the value obtained from such a measurement to the actual temperature of the heater adequately.

The completed implementation of the described solutions enabled the simulation of the thermal storage's operation under realistic conditions. Several simulations were conducted for various heating schedules. Each of them was applied for a period of 168 h (7 days), during which the change in the average temperature of the sand bed and the

temperature of the heaters, as well as the moments of their start-up and shutdown, were observed. The variants were collected and described in Table 9, and the heating of the sand bed was presented in Fig. 17. It is worth noting that the simulation for constant heating for a minimum temperature of 883 K throughout exceeds the alpha to beta quartz phase transformation temperature. For this reason, this variant should be treated only in an illustrative manner. A lower specific heat of the storage above the phase transition temperature would most likely result in a lower value of stored energy. The rest of the analyzed cases, according to preliminary assumptions, reach temperatures higher than the phase transition only in a minor part of the volume.

The first analyses examined charging the storage in a scenario where the heaters were allowed to run continuously for the entire study period. However, the heater restart temperature changed throughout the analysis.

Changing the minimum temperature at which the heating element starts up primarily affected the final temperature of the charged sand bed, which deviated more and more from the setpoint in each subse-

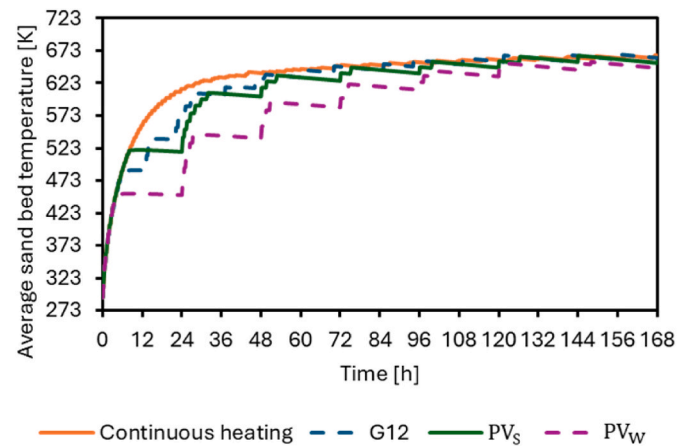


Fig. 20. Variation of average sand bed temperature for different charging schedule options.

Table 10

Comparison of the different schedules in terms of the time required to heat the sand bed to a given temperature value.

$T_{s,ave}$ goal [K]	Time to reach a given temperature [h]			
	G12	Continuous heating, $T_{res} = 673$ K	PV_S	PV_W
350	0.86	0.86	0.86	0.86
450	4.00	4.00	4.00	23.97
550	22.19	10.83	24.78	48.14
650	91.00	71.03	113.50	130.14

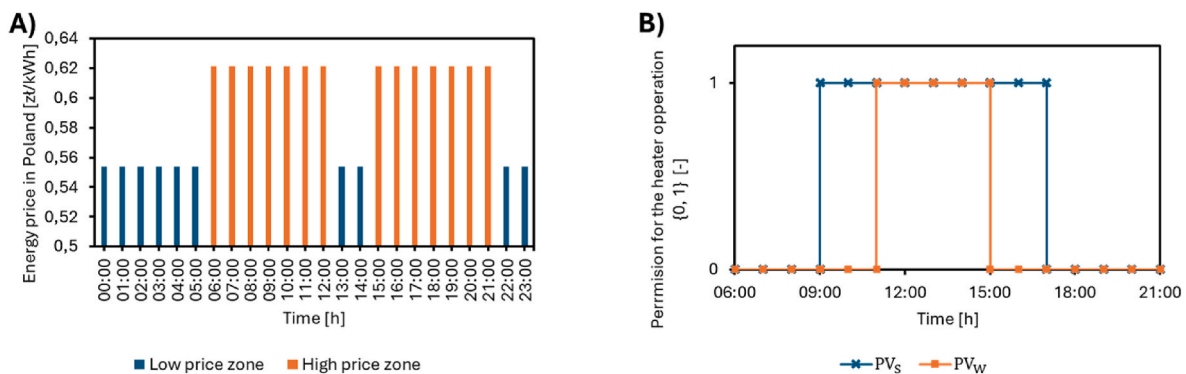


Fig. 19. Profiles created for analysis: A) daily energy price profile for G12 tariff in Poland, B) photovoltaic operation profile for a statistical summer and winter day.

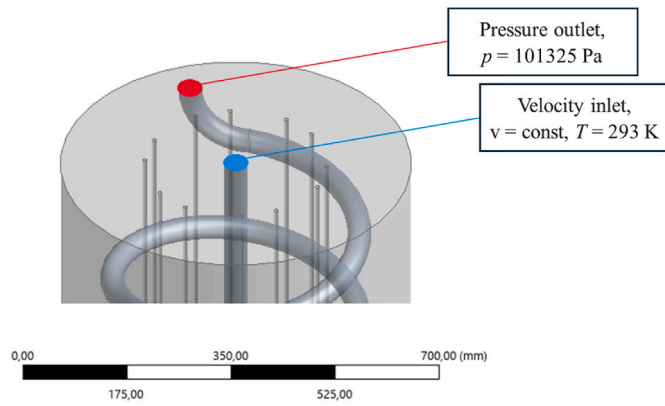


Fig. 21. Location of modified boundary conditions allowing fluid flow in piping.

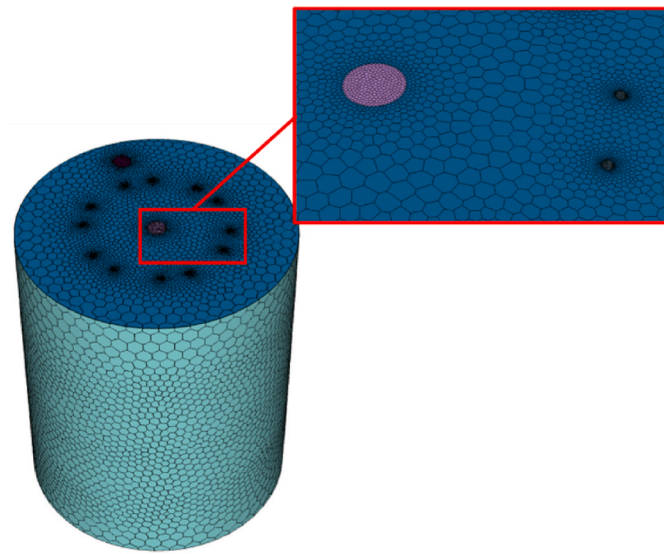


Fig. 22. Numerical mesh of modified model considering fluid flow in pipe heat exchanger.

Table 11

Mesh parameters of modified model considering fluid flow in pipe heat exchanger.

Parameter	Value	Allowed value according to ANSYS Workbench Manual documentation [41]
Number of nodes	24 768 070	–
Number of cells	4 929 294	–
Max/Min orthogonal quality	0.998/0.500	>0.05/~1
Avg orthogonal quality	0.953	>0.6
Min/Max skewness	0.00153/0.499	~0/<0.95
Avg skewness	0.0467	<0.4

quent variant. It is also worth focusing on the number of electric heater triggers. For the continuous heating variants with restart temperature (T_{res} [K]) of 673 and 773 K, it was a similar number, but for $T_{res} = 873$ K it increased significantly. The restart value of the heater temperature for this variant was so high that during the first 2 h of heating, it starts up more often than once every 8 min (for T_{res} of 673 K and 773 K it restarted once every 30 and 17 min, respectively). Frequent activation of heaters

Table 12

Results of variant analysis for different values of flow rate through the piping during energy withdrawal from sand bed.

Velocity on inlet [m/s]	Mass flux [kg/h]	DR [-]	P_{max} [W]	$P_{\%2h}$ [%]	$P_{\%10h}$ [%]
0.5	2.64	0.88	280.49	0.92 %	6.40 %
0.75	3.97	0.91	417.28	2.60 %	11.39 %
1	5.29	0.93	551.78	5.62 %	17.63 %
2	10.58	0.95	1028.16	17.79 %	37.24 %
3	15.86	0.96	1425.42	24.24 %	47.69 %
5	26.44	0.97	2082.76	32.80 %	59.75 %

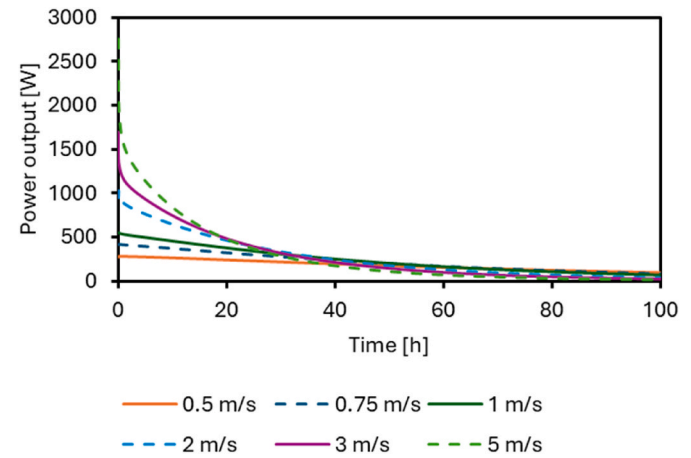


Fig. 23. Variation of power over time during storage discharge for different inlet velocities.

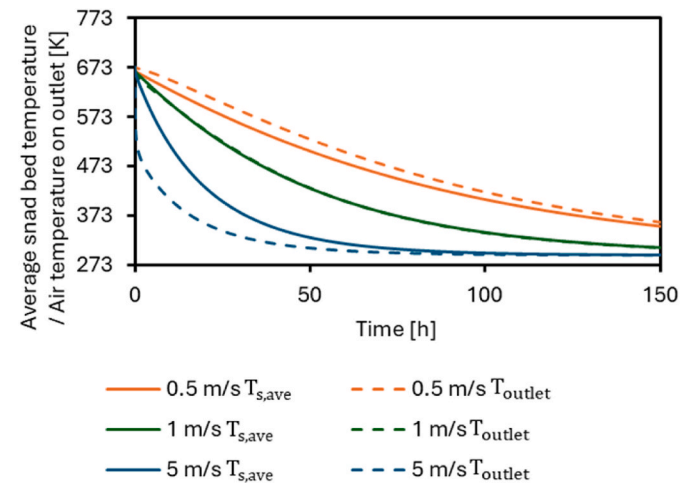


Fig. 24. Comparison of variations in average sand bed temperature and outlet air temperature for different inlet flow velocity values.

can have a negative impact on the rate of degradation of the contactor or other device used to operate the heater in question. This fact should be taken into consideration when designing a similar system. To illustrate the variation in the frequency of heater triggering, Fig. 18 shows the average temperature of the sand bed and one of the heaters over time.

Based on the presented relationships, as the temperature difference between the heaters and the sand bed decreases, the frequency of heater actuation also decreases. Importantly, in the case of a lower T_{res} , the frequency is significantly lower throughout the entire analyzed period. Table 9 also presents the calculated energy density values. To compare them with other types of heat storage, the constant charging variant for

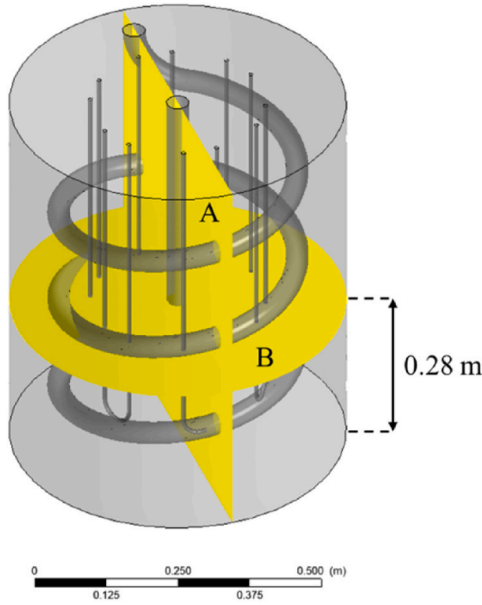


Fig. 25. Location of cross sections from which temperature distributions for discharge analysis were taken. A – vertical plane, B – horizontal plane.

$T_{res} = 883$ K was selected. This variant demonstrates the full potential of the proposed system. The energy density values can be compared to an alternative heat storage method – a sensible heat storage system based on water. This type of storage was chosen due to its widespread popularity. Assuming the maximum possible temperature to which water can be heated in a tank is 363 K, the energy density of such a storage system can be easily determined. Assuming a constant specific heat capacity of $C_p = 4200$ J/kgK, a density of $\rho = 1000$ kg/m³, and a reference temperature analogous to the simulations (293 K), the energy density is 394 MJ/m³ or 81.7 kWh/m³, which is nearly three times lower than the value obtained for the sand bed. This fact highlights the significant potential of utilizing this technology.

The G12 variant refers to the name of the electricity purchase tariff available from Polish energy distributors. This is a tariff based on two or three values of energy prices for particular hours of the day. An example of the distribution of purchase prices used in the following analysis is shown in Fig. 19. It is worth noting that the amounts shown in the chart are illustrative and, except for highlighting their differences, will not be relevant to the analysis. The schedule was imported into the model in a way that allowed the heaters to operate only during periods of low electricity purchase price. In addition to the schedule corresponding to the G12 tariff, schedules were created for the operation of the studied storage in cooperation with the photovoltaic installation. One corresponds to a statistical summer day (PV_S), and the other to a winter day (PV_W). These schedules are shown in Fig. 19. They were introduced into the model in an analogous way to the G12 tariff, allowing the heater to operate only when there was sufficient solar energy to power it.

Completed models with implemented operating schedules were used to simulate the change in the average temperature of the sand bed and the heaters. Both the schedule based on the G12 tariff and the integration with the PV installation assumed a $T_{res} = 673$ K. Due to this fact, the G12, PV_S, and PV_W, along with continuous heating $T_{res} = 673$ K variants, were compared with each other (see Fig. 20).

Each proposed charging schedule can heat the sand bed to ~650 K or more within 7 days. The schedule characterized by larger intervals during charging prevents the heater from warming up closer to the minimum temperature due to energy losses to the environment. Additionally, each of the analyzed variants exhibits distinct charging rates. To compare these cases, Table 10 summarizes the times required for the sand to reach an average temperature of 350, 450, 550, and 650 K.

The data in Table 9 shows that the differences between the SR for continuous heating, G12, and PV_S and PV_W are small. The largest discrepancies are observed here in charging times (see Table 10). The implemented control logic closely relates the storage operation to the temperature of the heaters. This shows some limitation on the maximum amount of energy that can be stored in one day. In future research, the focus should be on finding optimal intervals between charging cycles. A certain solution to this problem could be the creation of an array of smaller thermal storage units, which would be charged sequentially. It would be possible to maintain a continuous intake of excess energy while avoiding overheating of the heaters.

7. Discharge of thermal storage

Previous analyses have shown that the sand bed can be heated with electric heaters on various operating schedules. Further studies were conducted to evaluate the possibility of extracting the stored heat. In this case, the model required further modifications. In the new model, the fluid present in the piping flowed through it in a constant mass flux (see Fig. 21).

The consideration of fluid flow in the model affected the governing equations used in the ANSYS Fluent solver. The equation representing the principle of energy conservation needed to be modified to consider the buoyancy term, thus implementing the substance derivative (Eq. (13)). Additionally, the equations of continuity (Eq. (14)) and momentum (Eqs. (15)–(17)) had to be included. The airflow in the piping, regardless of the medium's analyzed velocity, was turbulent. The k - ϵ (Eqs. (18) and (19)) model was used to calculate the flow near the exchanger walls [43].

$$\rho \frac{DT}{Dt} = \frac{k_c}{C_p} \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right) \quad (13)$$

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \vec{V} = 0 \quad (14)$$

$$\rho \frac{Du}{Dt} = -\frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) \quad (15)$$

$$\rho \frac{Dv}{Dt} = -\frac{\partial p}{\partial y} + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right) \quad (16)$$

$$\rho \frac{Dw}{Dt} = -\frac{\partial p}{\partial z} + \mu \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right) \quad (17)$$

$$\frac{Dk}{Dt} = \frac{G_k}{\rho} - \epsilon + \left(\frac{\nu_t}{\sigma_k} \text{grad}k \right) \quad (18)$$

$$\frac{D\epsilon}{Dt} = C_{1\epsilon} \frac{\epsilon}{k} \frac{G_k}{\rho} - C_{2\epsilon} \frac{\epsilon^2}{k} + \left(\frac{\nu_t}{\sigma_\epsilon} \text{grad}\epsilon \right) \quad (19)$$

where $C_{1\epsilon}$, $C_{2\epsilon}$ – empirical model constants [–]

In addition to modifications to the governing equations of the model, the numerical mesh for the fluid domain inside the sand bed also had to be modified. For this reason, the cells inside this volume were reduced to allow satisfactory y^+ values on the walls of the piping. For the k - ϵ model, these values should be in the range between 20 and 200 [41]. The finished mesh is shown in Fig. 22 and described in Table 11.

The boundary conditions were also changed. The air domain, previously surrounded solely by walls, was equipped with a velocity inlet and pressure outlet. The velocity inlet was described by a constant velocity and temperature, while the pressure outlet simulated air escaping from the domain to a constant atmospheric pressure.

The analysis focused on examining the impact of altering the velocity of the medium flowing through the piping on the parameters of the entire storage system. Six variants differing in the value of air mass flux

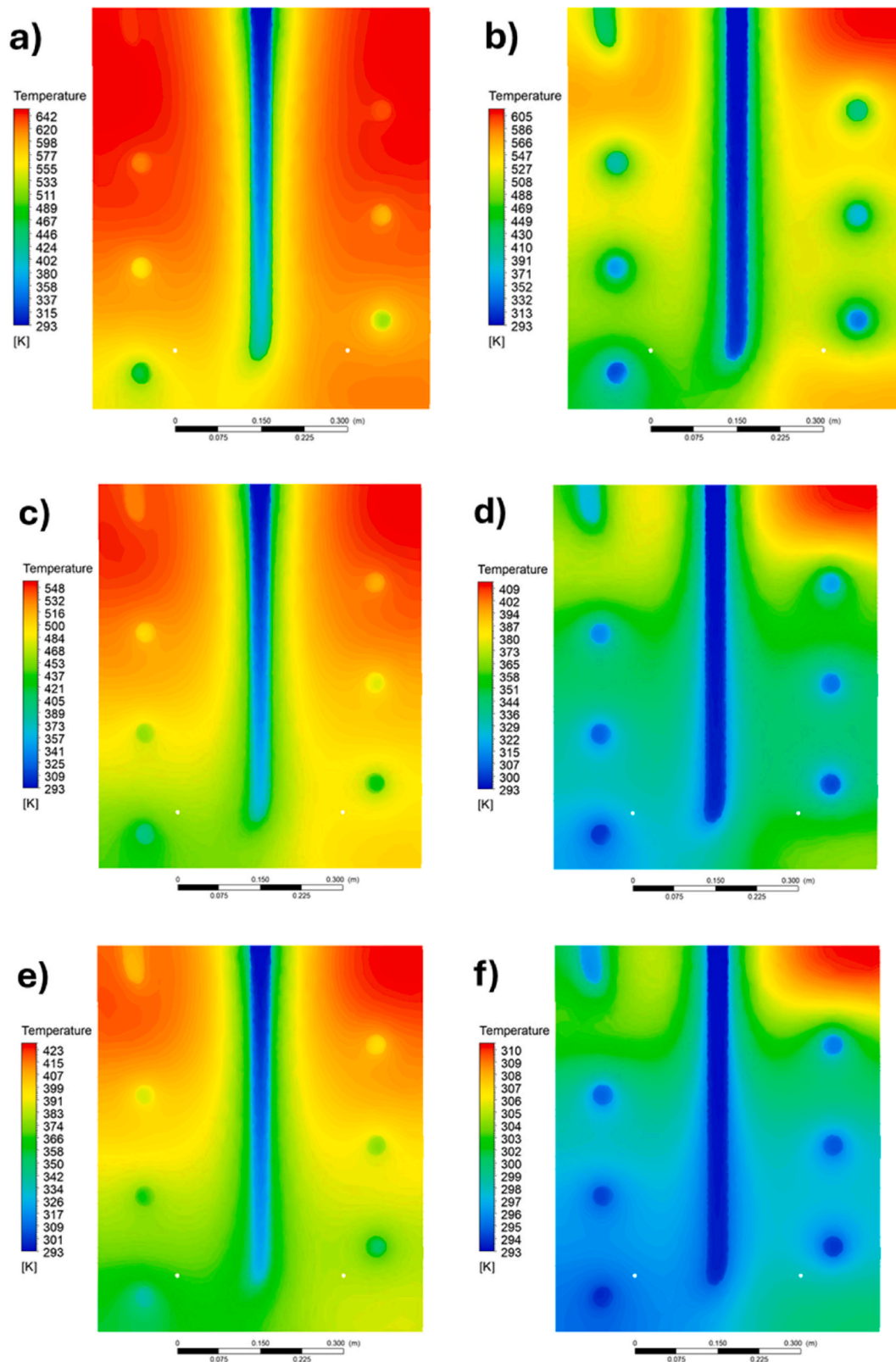


Fig. 26. Distribution of sand bed temperature for different time steps and flow velocities - cross section A. a) $t = 12$ h, $v = 0.5$ m/s, b) $t = 12$ h, $v = 3$ m/s, c) $t = 48$ h, $v = 0.5$ m/s, d) $t = 48$ h, $v = 3$ m/s, e) $t = 108$ h, $v = 0.5$ m/s, f) $t = 108$ h, $v = 3$ m/s.

were prepared. For each of them, a simulation lasted long enough for the bed to cool down to a temperature of ~ 300 K. The model's initial conditions coincided with the values obtained after the continuous heating simulation ($T_{res} = 673$ K). The airflow was calculated separately, and its

stabilized profile was initialized to match the heated model. The discharge ratio (DR) was introduced to compare the results, and it was calculated according to Eq. (20). Additionally, the maximum power of the airflow at the domain exit was considered, as well as its values after 2

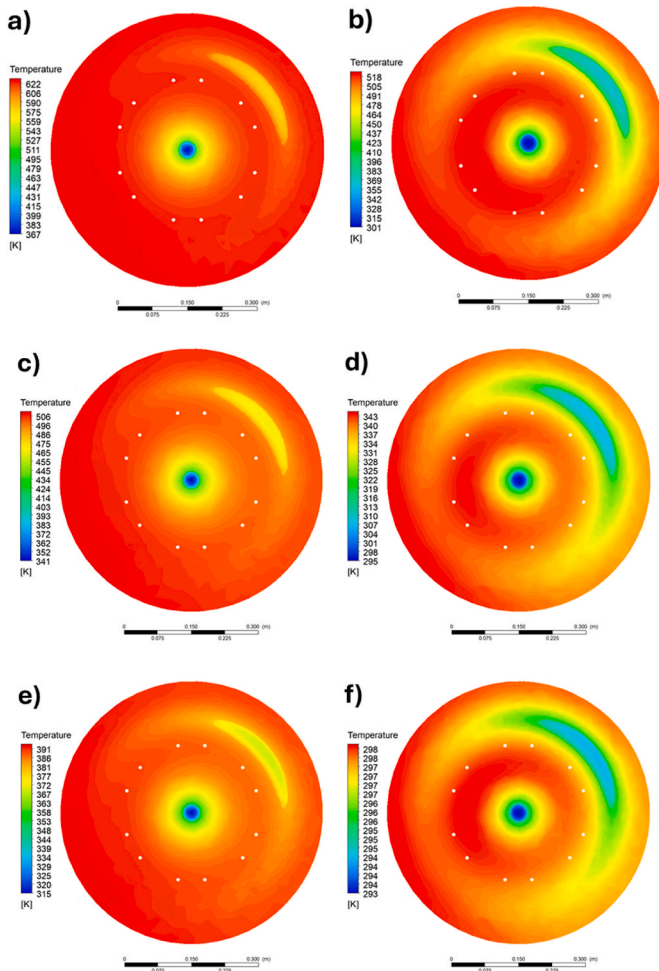


Fig. 27. Distribution of sand bed temperature for different time steps and flow velocities - cross section B. a) $t = 12$ h, $v = 0.5$ m/s, b) $t = 12$ h, $v = 3$ m/s, c) $t = 48$ h, $v = 0.5$ m/s, d) $t = 48$ h, $v = 3$ m/s, e) $t = 108$ h, $v = 0.5$ m/s, f) $t = 108$ h, $v = 3$ m/s.

and 10 h of simulation.

$$DR = \frac{E_{outlet}}{E_{outlet} + E_{loss}} \quad (20)$$

Where: E_{outlet} , E_{loss} - successively: the sum of the energy taken away by the airflow and the sum of the energy losses to the environment [MJ].

The results are gathered in Table 12. It shows that the maximum observed power (P_{max}) on output and DR increased with increasing air velocity. Since DR is determined by the amount of losses to the environment, increasing its value with mass flow rate is logical. Removing heat from the storage in a shorter time translates into a shorter period during which the storage can lose energy to the environment. Despite this, even the worst DR = 0.88 for $v = 0.5$ m/s should not be considered a bad result. In addition, Table 11 shows the power losses after 2 h ($P_{\%,2h}$) and 10 h ($P_{\%,10h}$) of operation for each case. The decrease is faster for higher flow rates, suggesting a critical examination of the maximum discharge power value. Future studies based on optimizing such a system should consider the appropriate flow rate selection to maximize the average power over a given period. Alternatively, the assumption of a constant flow rate could be replaced with the ability to control the power output through this parameter. To better illustrate the differences between the variants, graphs were created showing the time dependence of the power output of the outlet air flux (see Fig. 23) and the sand bed and outlet air temperatures (see Fig. 24). Cases (0.5, 1, and 5 m/s) were selected for this purpose.

What coincides with the data in Table 11 is that as the air velocity increases, an increase in instantaneous power can be observed in the first hours of the simulation ($t < 33$ h). The variant, assuming the lowest air velocity - 0.5 m/s, is characterized by the smoothest decrease in power at the outlet. Data from the graph allows to observe some convergence of the output power values for each variant. After about 33 h, the power ratings are very close to each other, with an average of ~ 257 W. Interestingly, after this point, the discrepancy between the variants increases, and the highest power is observed for the system with the lowest flow rate.

The graph showing the change in average sand and air temperature at the outlet have a similar character. The higher the air velocity, the steeper the temperature drop in the first hours of the simulation. It is worth noting that the average temperature of the bed and the air at the outlet for low velocities has a nature close to linear. As the speed of the fluid increases, the difference between the average temperature of the sand and air at the outlet increases. The graph shows that the temperature difference for both high and low flow rates at the beginning is characterized by the highest value, which decreases over time to zero. It is worth noting that for low flow velocity, the air temperature at the outlet during the entire analyzed period is higher than the average temperature of the sand bed. On the other hand, for the variant $v = 1$ m/s, these values practically overlap. This fact is related to the temperature distribution of the sand bed. When cold air flows through the inlet to the domain, it firstly cools the upper parts of the bed. As the medium flows through the storage, the temperature differences between the air and the sand become smaller and smaller. The flow velocity value of 1 m/s coincidentally is chosen so that the heat flux taken up by the air in different parts of the sand bed is equal to or at least close to the maximum heat flux that can be found in the sand. These relationships can be observed in the temperature distributions for the different variants (see Figs. 26 and 27) at cross-sections show in Fig. 25.

Presented contours directly indicate the differences between the low (0.5 m/s) and high (3 m/s) inlet air velocity variants. The temperature distribution for the high velocity variant changes rapidly throughout the process. Areas of much lower temperature around the piping can be observed. This phenomenon has already been described earlier when analyzing the graphs. The cooled area near the heat exchanger is the direct cause of limiting the maximum power of the airflow. Lower velocity of the medium results in a more homogeneous temperature distribution throughout the operating time. The strong influence of the low thermal conductivity of the sand bed becomes apparent here. Analyzing only the heat removal from the storage, increasing the thermal conductivity would positively affect the maximum power of the air flux. However, one should not forget the positive aspect of low conductivity, which reduces losses to the environment and thus increases the storage capacity of this design.

8. Conclusions

The research presented here demonstrated a methodology that allowed a multivariate analysis of the operation of heat storage using sand as a storage material. The cylindrical storage unit was based on quartz sand. The system assumed heating of the sand bed by electric heaters and heat removal through a pipe exchanger with air flowing through it. Several analyses were conducted to enhance the efficiency of the investigated system. The studies included the distribution of heaters inside the sand bed, the use of different charging schedules, and the effect of air velocity through the heat exchanger. Based on the above research, a number of conclusions can be drawn, as presented below.

1. The model was shown to be able to calculate the temperature distribution in the sand bed with acceptable accuracy. The developed model can be considered sufficient for variant analyses.
2. The appropriate distribution of heaters affects the operation of the storage. Smaller losses are observed for heaters closer to the axis of

the cylinder, while their temperature depends on their proximity to each other and the insulation layer.

3. The use of sand storage heating schedules will work best with RES systems (time-unstable sources) or variable tariffs. Analyses, however, have demonstrated some versatility in the use of the proposed system compared to other possible schedules.
4. Higher air flow rates in the exchanger lead to faster heat removal and higher instantaneous power at the outlet. At the same time, however, too high a flow rate causes a more rapid local cooling of the sand near the exchanger pipes, which, after an initial increase in power, leads to a faster decline. As a result, there is a trade-off between maximum heat-taking power and the stability and duration of the discharge process.

CRediT authorship contribution statement

Maksymilian Homa: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Krzysztof Sornek:** Writing – original draft, Supervision, Investigation, Conceptualization. **Wojciech Goryl:** Writing – original draft, Supervision, Investigation, Conceptualization. **Karolina Papis-Frączek:** Validation, Software, Methodology. **Paweł Żak:** Writing – review & editing, Supervision, Methodology, Data curation. **Rafał Dańko:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was carried out under Subvention no. 16.16.210.476 from the Faculty of Energy and Fuels and Subvention no. 16.16.170.654/B507 from the Faculty of Foundry Engineering, AGH University of Krakow. The research project was partly supported by the „Excellence initiative – research university” program for the AGH University of Krakow.

Data availability

The authors are unable or have chosen not to specify which data has been used.

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